

Human review packet: GJ19OptimalBinaryRatingSystems

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: GJ19OptimalBinaryRatingSystems
Paper id: GJ19OptimalBinaryRatingSystems
Source version: not recorded
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Rows: 59
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Source URL: [official source](#)

Review status: 0/59 source-claim screens; 0/62 library prerequisite screens. This is a review worksheet while those direct checks remain pending; it is not a final validation receipt.
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How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) EconCSLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-EconCSLib-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper GJ19OptimalBinaryRatingSystems --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's reviewer-owned review record.

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Material library prerequisites

EconCSLib.Optimization.IsLexicographicMaximizerOn

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{\alpha : \text{Type } u_1\} \text{ (feasible : } \alpha \rightarrow \text{Prop) (primary secondary : } \alpha \rightarrow \mathbb{R}) (x : \alpha),$
 $\text{feasible } x \wedge \forall (y : \alpha), \text{feasible } y \rightarrow \text{primary } y < \text{primary } x \vee \text{primary } y = \text{primary } x \wedge \text{secondary } y \leq \text{secondary } x$

Exact Lean library declaration

```
def IsLexicographicMaximizerOn {α : Type*} (feasible : α → Prop)
  (primary secondary : α → ℝ) (x : α) : Prop :=
  feasible x ∧ ∀ y, feasible y →
    primary y < primary x ∨
      (primary y = primary x ∧ secondary y ≤ secondary x)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.IsMaximizerOn

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {α : Type u_1} (feasible : α → Prop) (objective : α → ℝ) (x : α),  
  feasible x ∧ ∀ (y : α), feasible y → objective y ≤ objective x
```

Exact Lean library declaration

```
def IsMaximizerOn {α : Type*} (feasible : α → Prop)  
  (objective : α → ℝ) (x : α) : Prop :=  
  feasible x ∧ ∀ y, feasible y → objective y ≤ objective x
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.IsMinimizerOn

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{\alpha : \text{Type } u_1\} \text{ (feasible : } \alpha \rightarrow \text{Prop) (objective : } \alpha \rightarrow \mathbb{R}) (x : \alpha),$
 $\text{feasible } x \wedge \forall (y : \alpha), \text{feasible } y \rightarrow \text{objective } x \leq \text{objective } y$

Exact Lean library declaration

```
def IsMinimizerOn {α : Type*} (feasible : α → Prop)
  (objective : α → ℝ) (x : α) : Prop :=
  feasible x ∧ ∀ y, feasible y → objective x ≤ objective y
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.nestedBisectionStepBound

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(M \ L : \mathbb{N}) \rightarrow (L + 1) * (1 + (M - 3) * L)$

Exact Lean library declaration

```
def nestedBisectionStepBound (M L : ℕ) : ℕ :=  
  (L + 1) * (1 + (M - 3) * L)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.realBisectionMidpoint

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\text{lower upper} : \mathbb{R}) \rightarrow (\text{lower} + \text{upper}) / 2$

Exact Lean library declaration

```
noncomputable def realBisectionMidpoint (lower upper : ℝ) : ℝ :=  
  (lower + upper) / 2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.realBisectionStep

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(above : ℝ → Bool) →  
(lower upper : ℝ) →  
  have mid := EconCSLib.Optimization.realBisectionMidpoint lower upper;  
  if above mid = true then (lower, mid) else (mid, upper)
```

Exact Lean library declaration

```
noncomputable def realBisectionStep  
  (above : ℝ → Bool) (lower upper : ℝ) : ℝ × ℝ :=  
  let mid := realBisectionMidpoint lower upper  
  if above mid then (lower, mid) else (mid, upper)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.realBisectionStepFn

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(above : $\mathbb{R} \rightarrow \text{Bool}$) $\rightarrow$ (p : $\mathbb{R} \times \mathbb{R}$) $\rightarrow$ EconCSLib.Optimization.realBisectionStep above p.1 p.2

Exact Lean library declaration

```
noncomputable def realBisectionStepFn
  (above :  $\mathbb{R} \rightarrow \text{Bool}$ ) (p :  $\mathbb{R} \times \mathbb{R}$ ) :  $\mathbb{R} \times \mathbb{R}$  :=
  realBisectionStep above p.1 p.2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.realBisectionRun

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(above : $\mathbb{R} \rightarrow \text{Bool}$) $\rightarrow$ (n : $\mathbb{N}$) $\rightarrow$ (lower upper : $\mathbb{R}$) $\rightarrow$ (EconCSLib.Optimization.realBisectionStepFn above)^[n] (lower, upper)

Exact Lean library declaration

```
noncomputable def realBisectionRun
  (above :  $\mathbb{R} \rightarrow \text{Bool}$ ) (n :  $\mathbb{N}$ ) (lower upper :  $\mathbb{R}$ ) :  $\mathbb{R} \times \mathbb{R}$  :=
  Nat.iterate (realBisectionStepFn above) n (lower, upper)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.ExponentialRateCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$(\mathbb{N} \rightarrow \mathbb{R}) \rightarrow \mathbb{R} \rightarrow \text{Prop}$

Exact Lean library declaration

```
structure ExponentialRateCertificate (p :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (rate :  $\mathbb{R}$ ) where
  eventually_pos :  $\forall^f n \text{ in } \text{atTop}, \theta < p\ n$ 
  has_rate : HasExponentialRate p rate
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteMeasurableSetPartition

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{α : Type u_1} →  
  [inst : MeasurableSpace α] → MeasureTheory.Measure α → (Piece : Type u_2) → [Fintype Piece] → Type (max u_1 u_2)
```

Exact Lean library declaration

```
structure FiniteMeasurableSetPartition  
  {α : Type*} [MeasurableSpace α]  
  (μ : Measure α) (Piece : Type*) [Fintype Piece] where  
  support : Set α  
  pieceSet : Piece → Set α  
  measurable_piece : ∀ piece, MeasurableSet (pieceSet piece)  
  pairwiseDisjoint :  
    Pairwise (fun piece₁ piece₂ => Disjoint (pieceSet piece₁) (pieceSet piece₂))  
  cover : Set.iUnion pieceSet = support
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteMeasurableSetPartition.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{α : Type u_1} →
  [inst : MeasurableSpace α] →
  {μ : MeasureTheory.Measure α} →
  {Piece : Type u_2} →
  [inst_1 : Fintype Piece] →
  (support : Set α) →
  (pieceSet : Piece → Set α) →
  (∀ (piece : Piece), MeasurableSet (pieceSet piece)) →
  (Pairwise fun piece₁ piece₂ => Disjoint (pieceSet piece₁) (pieceSet piece₂)) →
  Set.iUnion pieceSet = support → EconCSLib.Probability.FiniteMeasurableSetPartition μ Piece
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteMeasurableSetPartition.orderedRealIocNatCutpoints

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(μ : MeasureTheory.Measure ℝ) →
(n : ℕ) →
(cut : ℕ → ℝ) →
(hmono : Monotone cut) →
{ support := Set.Ioc (cut 0) (cut n), pieceSet := fun piece => Set.Ioc (cut ↑piece) (cut (↑piece + 1)),
  measurable_piece := ⋯, pairwiseDisjoint := ⋯, cover := ⋯ }
```

Exact Lean library declaration

```
def orderedRealIocNatCutpoints
  (μ : Measure ℝ) (n : ℕ) (cut : ℕ → ℝ) (hmono : Monotone cut) :
  FiniteMeasurableSetPartition μ (Fin n) where
support := Set.Ioc (cut 0) (cut n)
pieceSet := fun piece => Set.Ioc (cut piece.val) (cut (piece.val + 1))
measurable_piece := by
  intro _piece
  exact measurableSet_Ioc
pairwiseDisjoint := by
  intro piece1 piece2 hne
  by_cases hlt : piece1.val < piece2.val
  · exact Set.Ioc_disjoint_Ioc_of_le
    (hmono (Nat.succ_le_of_lt hlt))
  · have hval_ne : piece1.val ≠ piece2.val := by
      intro hval
      exact hne (Fin.ext hval)
    have hgt : piece2.val < piece1.val := by
      exact lt_of_le_of_ne (Nat.le_of_not_gt hlt) hval_ne.symm
    exact (Set.Ioc_disjoint_Ioc_of_le
      (hmono (Nat.succ_le_of_lt hgt))).symm
cover := by
  ext x
  constructor
  · intro hx
    rcases Set.mem_iUnion.mp hx with ⟨piece, hxpiece⟩
    exact
      (lt_of_le_of_lt (hmono (Nat.zero_le piece.val)) hxpiece.1,
        le_trans hxpiece.2 (hmono (Nat.succ_le_of_lt piece.isLt)))
  · intro hx
    have hxUnion :
      x ∈ ⋃ i ∈ Finset.range n,
        Set.Ioc (cut i) (cut (i + 1)) :=
      Ioc_subset_biUnion_Ioc n cut hx
    rcases Set.mem_iUnion.mp hxUnion with ⟨i, hxi⟩
    rcases Set.mem_iUnion.mp hxi with ⟨hirange, hxpiece⟩
    have hi : i < n := by
      simpa using hirange
    exact Set.mem_iUnion.mpr ⟨⟨i, hi⟩, by simpa using hxpiece⟩
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteMeasurableSetPartition.orderedRealIocNatCutpointsMidpoint

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(cut : ℕ → ℝ) → {n : ℕ} → (piece : Fin n) → (cut ↑piece + cut (↑piece + 1)) / 2
```

Exact Lean library declaration

```
def orderedRealIocNatCutpointsMidpoint  
  (cut : ℕ → ℝ) {n : ℕ} (piece : Fin n) : ℝ :=  
  (cut piece.val + cut (piece.val + 1)) / 2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteMeasurableSetPartition.orderedRealIocNatCutpointsProdMidpoint

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(cut1 cut2 :  $\mathbb{N} \rightarrow \mathbb{R}$ ) →  
  {n m :  $\mathbb{N}$ } →  
    (piece : Fin n × Fin m) →  
      (EconCSLib.Probability.FiniteMeasurableSetPartition.orderedRealIocNatCutpointsMidpoint cut1 piece.1,  
       EconCSLib.Probability.FiniteMeasurableSetPartition.orderedRealIocNatCutpointsMidpoint cut2 piece.2)
```

Exact Lean library declaration

```
def orderedRealIocNatCutpointsProdMidpoint  
  (cut1 cut2 :  $\mathbb{N} \rightarrow \mathbb{R}$ ) {n m :  $\mathbb{N}$ } (piece : Fin n × Fin m) :  
     $\mathbb{R} \times \mathbb{R}$  :=  
  (orderedRealIocNatCutpointsMidpoint cut1 piece.1,  
   orderedRealIocNatCutpointsMidpoint cut2 piece.2)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteMeasurableSetPartition.pieceSet

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {α : Type u_1} [inst : MeasurableSpace α] {μ : MeasureTheory.Measure α} {Piece : Type u_2} [inst_1 : Fintype Piece]
  (self : EconCSLib.Probability.FiniteMeasurableSetPartition μ Piece) (a : Piece) (a_1 : α), self.2 a a_1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteMeasurableSetPartition.piecewiseConstKernel

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →
{Piece : Type u_2} →
[inst : MeasurableSpace α] →
[inst_1 : Fintype Piece] →
{μ : MeasureTheory.Measure α} →
(P : EconCSLib.Probability.FiniteMeasurableSetPartition μ Piece) →
(p : Piece → ℕ → ℝ) → (n : ℕ) → (x : α) → ∑ piece, (P.pieceSet piece).indicator (fun x => p piece n) x
```

Exact Lean library declaration

```
def piecewiseConstKernel
(P : FiniteMeasurableSetPartition μ Piece)
(p : Piece → ℕ → ℝ) (n : ℕ) (x : α) : ℝ :=
∑ piece : Piece, (P.pieceSet piece).indicator (fun _ : α => p piece n) x
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteMeasurableSetPartition.support

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {α : Type u_1} [inst : MeasurableSpace α] {μ : MeasureTheory.Measure α} {Piece : Type u_2} [inst_1 : Fintype Piece]
  (self : EconCSLib.Probability.FiniteMeasurableSetPartition μ Piece) (a : α), self.1 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteMeasurableSetPartition.prod

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →
{Piece : Type u_2} →
[inst : MeasurableSpace α] →
[inst_1 : Fintype Piece] →
{μ : MeasureTheory.Measure α} →
{β : Type u_4} →
{Pieceβ : Type u_5} →
[inst_2 : MeasurableSpace β] →
[inst_3 : Fintype Pieceβ] →
{ν : MeasureTheory.Measure β} →
(P : EconCSLib.Probability.FiniteMeasurableSetPartition μ Piece) →
(Q : EconCSLib.Probability.FiniteMeasurableSetPartition ν Pieceβ) →
{ support := P.support ×s Q.support,
  pieceSet := fun piece => P.pieceSet piece.1 ×s Q.pieceSet piece.2, measurable_piece := "",
  pairwiseDisjoint := "", cover := "" }
```

Exact Lean library declaration

```
def prod {β Pieceβ : Type*} [MeasurableSpace β] [Fintype Pieceβ]
  {ν : Measure β}
  (P : FiniteMeasurableSetPartition μ Piece)
  (Q : FiniteMeasurableSetPartition ν Pieceβ) :
  FiniteMeasurableSetPartition (μ.prod ν) (Piece × Pieceβ) where
support := P.support ×s Q.support
pieceSet := fun piece => P.pieceSet piece.1 ×s Q.pieceSet piece.2
measurable_piece := by
  intro piece
  exact (P.measurable_piece piece.1).prod (Q.measurable_piece piece.2)
pairwiseDisjoint := by
  intro piece1 piece2 hne
  rw [Set.disjoint_left]
  intro x hx1 hx2
  rcases x with (x1, x2)
  by_cases hfirst : piece1.1 = piece2.1
  · have hsecond : piece1.2 ≠ piece2.2 := by
    intro hsecond
    exact hne (Prod.ext hfirst hsecond)
  · exact
    (Set.disjoint_left.mp
      (Q.pairwiseDisjoint hsecond)) hx1.2 hx2.2
  · exact
    (Set.disjoint_left.mp
      (P.pairwiseDisjoint hfirst)) hx1.1 hx2.1
cover := by
  ext x
  rcases x with (x1, x2)
  constructor
  · intro hx
    rcases Set.mem_iUnion.mp hx with (piece, hxpiece)
    have hx1 : x1 ∈ Set.iUnion P.pieceSet :=
      Set.mem_iUnion.mpr (piece.1, hxpiece.1)
    have hx2 : x2 ∈ Set.iUnion Q.pieceSet :=
      Set.mem_iUnion.mpr (piece.2, hxpiece.2)
    exact (by simp [P.cover] using hx1, by simp [Q.cover] using hx2)
  · intro hx
    have hx1 : x1 ∈ Set.iUnion P.pieceSet := by
      simp [P.cover] using hx.1
    have hx2 : x2 ∈ Set.iUnion Q.pieceSet := by
      simp [Q.cover] using hx.2
    rcases Set.mem_iUnion.mp hx1 with (piece1, hxpiece1)
    rcases Set.mem_iUnion.mp hx2 with (piece2, hxpiece2)
    exact Set.mem_iUnion.mpr ((piece1, piece2), {hxpiece1, hxpiece2})
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteMeasurableSetPartition.subpartition

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →
{Piece : Type u_2} →
[inst : MeasurableSpace α] →
[inst_1 : Fintype Piece] →
{μ : MeasureTheory.Measure α} →
(P : EconCSLib.Probability.FiniteMeasurableSetPartition μ Piece) →
(selected : Piece → Prop) →
[inst_2 : DecidablePred selected] →
{ support := ⋃ piece, P.pieceSet ↑piece, pieceSet := fun piece => P.pieceSet ↑piece,
  measurable_piece := "", pairwiseDisjoint := "", cover := "" }
```

Exact Lean library declaration

```
def subpartition
(P : FiniteMeasurableSetPartition μ Piece)
(selected : Piece → Prop) [DecidablePred selected] :
FiniteMeasurableSetPartition μ {piece : Piece // selected piece} where
support := Set.iUnion (fun piece : {piece : Piece // selected piece} =>
P.pieceSet piece.val)
pieceSet := fun piece => P.pieceSet piece.val
measurable_piece := by
  intro piece
  exact P.measurable_piece piece.val
pairwiseDisjoint := by
  intro piece1 piece2 hne
  have hval : piece1.val ≠ piece2.val := by
    intro hval
    exact hne (Subtype.ext hval)
  exact P.pairwiseDisjoint hval
cover := rfl
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteMeasurableSetPartition.selectedProduct

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →
{Piece : Type u_2} →
[inst : MeasurableSpace α] →
[inst_1 : Fintype Piece] →
{μ : MeasureTheory.Measure α} →
{β : Type u_4} →
{Pieceβ : Type u_5} →
[inst_2 : MeasurableSpace β] →
[inst_3 : Fintype Pieceβ] →
{ν : MeasureTheory.Measure β} →
(P : EconCSLib.Probability.FiniteMeasurableSetPartition μ Piece) →
(Q : EconCSLib.Probability.FiniteMeasurableSetPartition ν Pieceβ) →
(selected : Piece × Pieceβ → Prop) →
[inst_4 : DecidablePred selected] → (P.prod Q).subpartition selected
```

Exact Lean library declaration

```
def selectedProduct {β Pieceβ : Type*} [MeasurableSpace β] [Fintype Pieceβ]
  {ν : Measure β}
  (P : FiniteMeasurableSetPartition μ Piece)
  (Q : FiniteMeasurableSetPartition ν Pieceβ)
  (selected : Piece × Pieceβ → Prop) [DecidablePred selected] :
  FiniteMeasurableSetPartition (μ.prod ν)
  {piece : Piece × Pieceβ // selected piece} :=
  (P.prod Q).subpartition selected
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type $u_2 \rightarrow (\text{Rating} : \text{Type } u_3) \rightarrow [\text{Fintype Rating}] \rightarrow [\text{DecidableEq Rating}] \rightarrow \text{Type } (\max u_2 u_3)$

Exact Lean library declaration

```
structure FiniteRatingLDPModel (θ Rating : Type*) [Fintype Rating]
  [DecidableEq Rating] where
  typeLaw : θ → PMF Rating
  score : Rating → ℝ
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel.score

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{θ : Type u_2} →  
{Rating : Type u_3} →  
  [inst : Fintype Rating] →  
    [inst_1 : DecidableEq Rating] →  
      (self : EconCSLib.Probability.FiniteRatingLDPModel θ Rating) → (a : Rating) → self.2 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel.typeLaw

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{θ : Type u_2} →  
  {Rating : Type u_3} →  
    [inst : Fintype Rating] →  
      [inst_1 : DecidableEq Rating] → (self : EconCSLib.Probability.FiniteRatingLDPModel θ Rating) → (a : θ) → self.1 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteMGF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
[inst : Fintype α] → (μ : PMF α) → (score : α → ℝ) → (z : ℝ) → ∑ a, (μ a).toReal * Real.exp (z * score a)
```

Exact Lean library declaration

```
def finiteMGF (μ : PMF α) (score : α → ℝ) (z : ℝ) : ℝ :=  
  ∑ a : α, (μ a).toReal * Real.exp (z * score a)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteLogMGF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
[inst : Fintype α] → (μ : PMF α) → (score : α → ℝ) → (z : ℝ) → Real.log (EconCSLib.Probability.finiteMGF μ score z)
```

Exact Lean library declaration

```
def finiteLogMGF (μ : PMF α) (score : α → ℝ) (z : ℝ) : ℝ :=  
  Real.log (finiteMGF μ score z)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel.logMGF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{θ : Type u_2} →  
{Rating : Type u_3} →  
  [inst : Fintype Rating] →  
    [inst_1 : DecidableEq Rating] →  
      (M : EconCSLib.Probability.FiniteRatingLDPModel θ Rating) →  
        (t : θ) → (z : ℝ) → EconCSLib.Probability.finiteLogMGF (M.typeLaw t) M.score z
```

Exact Lean library declaration

```
def logMGF (M : FiniteRatingLDPModel θ Rating) (t : θ) (z : ℝ) : ℝ :=  
  finiteLogMGF (M.typeLaw t) M.score z
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{θ : Type u_2} →  
  {Rating : Type u_3} →  
    [inst : Fintype Rating] →  
      [inst_1 : DecidableEq Rating] →  
        (θ → PMF Rating) → (Rating → ℝ) → EconCSLib.Probability.FiniteRatingLDPModel θ Rating
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.HasAEEssentialInfimum

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{\alpha : \text{Type } u_1\} [\text{inst} : \text{MeasurableSpace } \alpha] (\mu : \text{MeasureTheory.Measure } \alpha) (\text{phi} : \alpha \rightarrow \mathbb{R}) (\text{rate} : \mathbb{R}),$
 $(\forall^m (x : \alpha) \partial\mu, \text{rate} \leq \text{phi } x) \wedge \forall \varepsilon > 0, \exists s, \text{MeasurableSet } s \wedge 0 < \mu.\text{real } s \wedge \forall x \in s, \text{phi } x \leq \text{rate} + \varepsilon$

Exact Lean library declaration

```
def HasAEEssentialInfimum
  {α : Type*} [MeasurableSpace α]
  (μ : Measure α) (phi : α → ℝ) (rate : ℝ) : Prop :=
  (∀m x ∂μ, rate ≤ phi x) ∧
  ∀ ε > 0, ∃ s : Set α,
    MeasurableSet s ∧
    0 < μ.real s ∧
    ∀ x, x ∈ s → phi x ≤ rate + ε
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.logDecay

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(p : \mathbb{N} \rightarrow \mathbb{R}) \rightarrow (n : \mathbb{N}) \rightarrow -(1/n)^{-1} * \text{Real.log } (p \ n)$

Exact Lean library declaration

```
def logDecay (p :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (n :  $\mathbb{N}$ ) :  $\mathbb{R}$  :=  
  -((n :  $\mathbb{R}$ )-1) * Real.log (p n)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.HasExponentialRate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall (p : \mathbb{N} \rightarrow \mathbb{R}) (rate : \mathbb{R}), \text{Filter.Tendsto } (\text{EconCSLib.Probability.logDecay } p) \text{ Filter.atTop } (nhds \text{ rate})$

Exact Lean library declaration

```
def HasExponentialRate (p :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (rate :  $\mathbb{R}$ ) : Prop :=  
  Tendsto (logDecay p) atTop (nhds rate)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.HasPositiveWeightNearAEEssentialInfimum

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$$\forall \{\alpha : \text{Type } u_1\} [\text{inst} : \text{MeasurableSpace } \alpha] (\mu : \text{MeasureTheory.Measure } \alpha) (w \text{ phi} : \alpha \rightarrow \mathbb{R}) (\text{rate } \varepsilon : \mathbb{R}),$$
$$\varepsilon > 0 \rightarrow \exists s \text{ } c, \theta < \mu.\text{real } s \wedge \theta < c \wedge \forall^m (x : \alpha) \partial \mu.\text{restrict } s, c \leq w \ x \wedge \text{phi } x \leq \text{rate} + \varepsilon$$

Exact Lean library declaration

```
def HasPositiveWeightNearAEEssentialInfimum
  {α : Type*} [MeasurableSpace α]
  (μ : Measure α) (w phi : α → ℝ) (rate : ℝ) : Prop :=
  ∀ ε > 0, ∃ s : Set α, ∃ c : ℝ,
    θ < μ.real s ∧ θ < c ∧
    ∀m x ∂μ.restrict s, c ≤ w x ∧ phi x ≤ rate + ε
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PairwiseThresholdRateTopLdpCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Seller : Type u_1} →
{Rating : Type u_2} →
{Pair : Type u_3} →
  [inst : Fintype Rating] →
  [inst_1 : DecidableEq Rating] →
    EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating →
      (Seller → ℝ) → (Pair → Seller) → (Pair → Seller) → Type u_3
```

Exact Lean library declaration

```
structure PairwiseThresholdRateTopLdpCertificate
  {Seller Rating Pair : Type*} [Fintype Rating] [DecidableEq Rating]
  (M : FiniteRatingLDPMModel Seller Rating) (sampleRate : Seller → ℝ)
  (pairHi pairLo : Pair → Seller) where
  rate : Pair → ℝ
  threshold_rate_top_eq :
    ∀ p : Pair,
      pairwiseSellerThresholdRateTop M sampleRate (pairHi p) (pairLo p) =
        (rate p : WithTop ℝ)
  leftTail :
    ∀ p : Pair,
      ExponentialRateCertificate
        (twoSampleFloorScoreGapLeftTailProb M sampleRate (pairHi p) (pairLo p))
        (rate p)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.bernoulliFirstEndpointEqualSplit

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(pHi : \mathbb{R}) \rightarrow (1 - \sqrt{(1 - pHi)}) / 2$

Exact Lean library declaration

```
def bernoulliFirstEndpointEqualSplit (pHi : ℝ) : ℝ :=  
  (1 - Real.sqrt (1 - pHi)) / 2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.bernoulliInteriorFailureSplitNumerator

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(pLo\ pHi : \mathbb{R}) \rightarrow (\sqrt{(1 - pLo)} - \sqrt{(1 - pHi)})^2$

Exact Lean library declaration

```
def bernoulliInteriorFailureSplitNumerator (pLo pHi : ℝ) : ℝ :=  
  (Real.sqrt (1 - pLo) - Real.sqrt (1 - pHi)) ^ 2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.bernoulliInteriorSuccessSplitNumerator

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(pLo\ pHi : \mathbb{R}) \rightarrow (\sqrt{pHi} - \sqrt{pLo})^2$

Exact Lean library declaration

```
def bernoulliInteriorSuccessSplitNumerator (pLo pHi : ℝ) : ℝ :=  
  (Real.sqrt pHi - Real.sqrt pLo) ^ 2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.bernoulliInteriorSplitDenominator

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(pLo pHi : ℝ) →  
  EconCSLib.Probability.bernoulliInteriorFailureSplitNumerator pLo pHi +  
  EconCSLib.Probability.bernoulliInteriorSuccessSplitNumerator pLo pHi
```

Exact Lean library declaration

```
def bernoulliInteriorSplitDenominator (pLo pHi : ℝ) : ℝ :=  
  bernoulliInteriorFailureSplitNumerator pLo pHi +  
  bernoulliInteriorSuccessSplitNumerator pLo pHi
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.bernoulliInteriorEqualSplit

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(pLo pHi : ℝ) →  
  EconCSLib.Probability.bernoulliInteriorFailureSplitNumerator pLo pHi /  
  EconCSLib.Probability.bernoulliInteriorSplitDenominator pLo pHi
```

Exact Lean library declaration

```
def bernoulliInteriorEqualSplit (pLo pHi : ℝ) : ℝ :=  
  bernoulliInteriorFailureSplitNumerator pLo pHi /  
  bernoulliInteriorSplitDenominator pLo pHi
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.bernoulliKL

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(a b : ℝ) → a * Real.log (a / b) + (1 - a) * Real.log ((1 - a) / (1 - b))
```

Exact Lean library declaration

```
def bernoulliKL (a b : ℝ) : ℝ :=  
  a * Real.log (a / b) + (1 - a) * Real.log ((1 - a) / (1 - b))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.bernoulliLastEndpointEqualSplit

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(pLo : \mathbb{R}) \rightarrow (1 + \sqrt{pLo}) / 2$

Exact Lean library declaration

```
def bernoulliLastEndpointEqualSplit (pLo : ℝ) : ℝ :=  
  (1 + Real.sqrt pLo) / 2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.binaryRatingScore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(b : \text{Bool}) \rightarrow \text{if } b = \text{true} \text{ then } 1 \text{ else } 0$

Exact Lean library declaration

```
def binaryRatingScore (b : Bool) : ℝ :=  
  if b then 1 else 0
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteIdScoreSum

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{\alpha : \text{Type } u_1\} \rightarrow \{\iota : \text{Type } u_2\} \rightarrow [\text{inst} : \text{Fintype } \iota] \rightarrow (\text{score} : \alpha \rightarrow \mathbb{R}) \rightarrow (\text{sample} : \iota \rightarrow \alpha) \rightarrow \sum i, \text{score } (\text{sample } i)$

Exact Lean library declaration

```
def finiteIdScoreSum {ι : Type*} [Fintype ι]
  (score : α → ℝ) (sample : ι → α) : ℝ :=
  ∑ i : ι, score (sample i)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.floorSampleCount

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} → (sampleRate : Seller → ℝ) → (θ : Seller) → (k : ℕ) → [↑k * sampleRate θ]₊
```

Exact Lean library declaration

```
def floorSampleCount {Seller : Type*} (sampleRate : Seller → ℝ)
  (θ : Seller) (k : ℕ) : ℕ :=
  Nat.floor ((k : ℝ) * sampleRate θ)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.realBernoulliPMF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(p : \mathbb{R}) \rightarrow (hp0 : 0 \leq p) \rightarrow (hp1 : p \leq 1) \rightarrow \text{PMF.bernoulli } \langle p, hp0 \rangle \dots$

Exact Lean library declaration

```
def realBernoulliPMF (p : ℝ) (hp0 : 0 ≤ p) (hp1 : p ≤ 1) : PMF Bool :=
  PMF.bernoulli ⟨p, hp0⟩ (by
    change p ≤ (1 : ℝ)
    exact hp1)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.realBinaryRatingLDPMoDel

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{θ : Type u_1} →  
(successProb : θ → ℝ) →  
  (hprob0 : ∀ (t : θ), 0 ≤ successProb t) →  
    (hprob1 : ∀ (t : θ), successProb t ≤ 1) →  
      { typeLaw := fun t => EconCSLib.Probability.realBernoulliPMF (successProb t) ... ...,  
        score := EconCSLib.Probability.binaryRatingScore }
```

Exact Lean library declaration

```
def realBinaryRatingLDPMoDel {θ : Type*}  
  (successProb : θ → ℝ)  
  (hprob0 : ∀ t, 0 ≤ successProb t)  
  (hprob1 : ∀ t, successProb t ≤ 1) :  
  FiniteRatingLDPMoDel θ Bool where  
  typeLaw := fun t => realBernoulliPMF (successProb t) (hprob0 t) (hprob1 t)  
  score := binaryRatingScore
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoBernoulliThresholdRate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(gHi\ gLo\ tHi\ tLo\ a : \mathbb{R}) \rightarrow gHi * EconCSLib.Probability.bernoulliKL\ a\ tHi + gLo * EconCSLib.Probability.bernoulliKL\ a\ tLo$

Exact Lean library declaration

```
def twoBernoulliThresholdRate
  (gHi gLo tHi tLo a : ℝ) : ℝ :=
  gHi * bernoulliKL a tHi + gLo * bernoulliKL a tLo
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfProd

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
{β : Type u_2} →  
  [inst : Fintype α] → [inst_1 : Fintype β] → (μ : PMF α) → (ν : PMF β) → PMF.ofFintype (fun p => μ p.1 * ν p.2) ...
```

Exact Lean library declaration

```
noncomputable def pmfProd {α β : Type*} [Fintype α] [Fintype β]  
  (μ : PMF α) (ν : PMF β) : PMF (α × β) :=  
  PMF.ofFintype (fun p : α × β => μ p.1 * ν p.2) (by  
    classical  
    have hμ : ∑ a : α, μ a = 1 := by  
      rw [← PMF.tsum_coe μ, tsum_fintype]  
    have hν : ∑ b : β, ν b = 1 := by  
      rw [← PMF.tsum_coe ν, tsum_fintype]  
    calc  
      ∑ p : α × β, μ p.1 * ν p.2  
      = ∑ a : α, ∑ b : β, μ a * ν b := by  
        rw [Fintype.sum_prod_type]  
      = ∑ a : α, μ a * (∑ b : β, ν b) := by  
        simp [Finset.mul_sum]  
      = 1 := by  
        simp [hμ, hν])
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfProduct

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(ι : Type u_1) →
(α : Type u_2) →
[inst : Fintype ι] →
[inst_1 : DecidableEq ι] → [inst_2 : Fintype α] → (μ : PMF α) → PMF.ofFintype (fun f => [] i, μ (f i)) ...
```

Exact Lean library declaration

```
noncomputable def pmfProduct (ι α : Type*) [Fintype ι] [DecidableEq ι] [Fintype α]
  (μ : PMF α) : PMF (ι → α) :=
  PMF.ofFintype (fun f : ι → α => [] i : ι, μ (f i)) (by
    classical
    have hcoord : ∑ a : α, μ a = 1 := by
      rw [← PMF.tsum_coe μ, tsum_fintype]
    calc
      ∑ f : ι → α, [] i : ι, μ (f i)
        = [] i : ι, ∑ a : α, μ a := by
          symm
          simp using
            (Finset.prod_univ_sum
              (t := fun _i : ι => (Finset.univ : Finset α))
              (f := fun _i a => μ a))
      _ = 1 := by simp [hcoord])
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleRatingLaw

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) →
(hi lo : Seller) →
(nHi nLo : ℕ) →
  EconCSLib.pmfProd (EconCSLib.pmfProduct (Fin nHi) Rating (M.typeLaw hi))
    (EconCSLib.pmfProduct (Fin nLo) Rating (M.typeLaw lo))
```

Exact Lean library declaration

```
noncomputable def twoSampleRatingLaw
  {Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
  (M : FiniteRatingLDPModel Seller Rating) (hi lo : Seller)
  (nHi nLo : ℕ) :
  PMF ((Fin nHi → Rating) × (Fin nLo → Rating)) :=
  EconCSLib.pmfProd
    (EconCSLib.pmfProduct (Fin nHi) Rating (M.typeLaw hi))
    (EconCSLib.pmfProduct (Fin nLo) Rating (M.typeLaw lo))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleScoreGapSum

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) →
{nHi nLo : ℕ} →
(cHi cLo : ℝ) →
(sample : (Fin nHi → Rating) × (Fin nLo → Rating)) →
cHi * EconCSLib.Probability.finiteIidScoreSum M.score sample.1 -
cLo * EconCSLib.Probability.finiteIidScoreSum M.score sample.2
```

Exact Lean library declaration

```
def twoSampleScoreGapSum
{Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
(M : FiniteRatingLDPModel Seller Rating) {nHi nLo : ℕ}
(cHi cLo : ℝ)
(sample : (Fin nHi → Rating) × (Fin nLo → Rating)) : ℝ :=
cHi * finiteIidScoreSum M.score sample.1 -
cLo * finiteIidScoreSum M.score sample.2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfExp

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{\alpha : \text{Type } u_1\} \rightarrow [\text{inst} : \text{Fintype } \alpha] \rightarrow [\text{DecidableEq } \alpha] \rightarrow (\mu : \text{PMF } \alpha) \rightarrow (f : \alpha \rightarrow \mathbb{R}) \rightarrow \sum a, (\mu a).toReal * f a$

Exact Lean library declaration

```
noncomputable def pmfExp {α : Type*} [Fintype α] [DecidableEq α]
  (μ : PMF α) (f : α → ℝ) : ℝ :=
  ∑ a, (μ a).toReal * f a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
  [inst : Fintype α] →  
    [inst_1 : DecidableEq α] →  
      (μ : PMF α) → (p : α → Prop) → [inst_2 : DecidablePred p] → EconCSLib.pmfExp μ fun a => if p a then 1 else 0
```

Exact Lean library declaration

```
noncomputable def pmfProb {α : Type*} [Fintype α] [DecidableEq α]  
  (μ : PMF α) (p : α → Prop) [DecidablePred p] : ℝ :=  
  pmfExp μ (fun a => if p a then 1 else 0)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleScoreGapStrictLeftProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating) →
(hi lo : Seller) →
(nHi nLo : ℕ) →
(cHi cLo : ℝ) →
EconCSLib.pmfProb (EconCSLib.Probability.twoSampleRatingLaw M hi lo nHi nLo) fun sample =>
EconCSLib.Probability.twoSampleScoreGapSum M cHi cLo sample < 0
```

Exact Lean library declaration

```
def twoSampleScoreGapStrictLeftProb
{Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
(M : FiniteRatingLDPMModel Seller Rating) (hi lo : Seller)
(nHi nLo : ℕ) (cHi cLo : ℝ) : ℝ :=
EconCSLib.pmfProb (twoSampleRatingLaw M hi lo nHi nLo)
(fun sample => twoSampleScoreGapSum M cHi cLo sample < 0)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleScoreGapTieProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) →
(hi lo : Seller) →
(nHi nLo : ℕ) →
(cHi cLo : ℝ) →
EconCSLib.pmfProb (EconCSLib.Probability.twoSampleRatingLaw M hi lo nHi nLo) fun sample =>
EconCSLib.Probability.twoSampleScoreGapSum M cHi cLo sample = 0
```

Exact Lean library declaration

```
def twoSampleScoreGapTieProb
{Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
(M : FiniteRatingLDPModel Seller Rating) (hi lo : Seller)
(nHi nLo : ℕ) (cHi cLo : ℝ) : ℝ :=
EconCSLib.pmfProb (twoSampleRatingLaw M hi lo nHi nLo)
(fun sample => twoSampleScoreGapSum M cHi cLo sample = 0)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSamplePkComplementErrorProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) →
(hi lo : Seller) →
(nHi nLo : ℕ) →
(cHi cLo : ℝ) →
2 * EconCSLib.Probability.twoSampleScoreGapStrictLeftProb M hi lo nHi nLo cHi cLo +
EconCSLib.Probability.twoSampleScoreGapTieProb M hi lo nHi nLo cHi cLo
```

Exact Lean library declaration

```
def twoSamplePkComplementErrorProb
{Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
(M : FiniteRatingLDPModel Seller Rating) (hi lo : Seller)
(nHi nLo : ℕ) (cHi cLo : ℝ) : ℝ :=
2 * twoSampleScoreGapStrictLeftProb M hi lo nHi nLo cHi cLo +
twoSampleScoreGapTieProb M hi lo nHi nLo cHi cLo
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleFloorPkComplementErrorProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) →
(sampleRate : Seller → ℝ) →
(hi lo : Seller) →
(k : ℕ) →
  have nHi := EconCSLib.Probability.floorSampleCount sampleRate hi k;
  have nLo := EconCSLib.Probability.floorSampleCount sampleRate lo k;
  EconCSLib.Probability.twoSamplePkComplementErrorProb M hi lo nHi nLo (↑nHi)-1 (↑nLo)-1
```

Exact Lean library declaration

```
def twoSampleFloorPkComplementErrorProb
  {Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
  (M : FiniteRatingLDPModel Seller Rating) (sampleRate : Seller → ℝ)
  (hi lo : Seller) (k : ℕ) : ℝ :=
  let nHi := floorSampleCount sampleRate hi k
  let nLo := floorSampleCount sampleRate lo k
  twoSamplePkComplementErrorProb M hi lo nHi nLo
  ((nHi : ℝ)-1) ((nLo : ℝ)-1)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.weightedBernoulliFailureBase

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(gHi \ gLo \ pHi \ pLo : \mathbb{R}) \rightarrow (1 - pLo) ^ (gLo / (gHi + gLo)) * (1 - pHi) ^ (gHi / (gHi + gLo))$

Exact Lean library declaration

```
def weightedBernoulliFailureBase (gHi gLo pHi pLo : ℝ) : ℝ :=  
  (1 - pLo) ^ (gLo / (gHi + gLo)) *  
  (1 - pHi) ^ (gHi / (gHi + gLo))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.weightedBernoulliSuccessBase

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(gHi \ gLo \ pHi \ pLo : \mathbb{R}) \rightarrow pLo \wedge (gLo / (gHi + gLo)) * pHi \wedge (gHi / (gHi + gLo))$

Exact Lean library declaration

```
def weightedBernoulliSuccessBase (gHi gLo pHi pLo : ℝ) : ℝ :=  
  pLo ^ (gLo / (gHi + gLo)) *  
  pHi ^ (gHi / (gHi + gLo))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.weightedBernoulliClosedRateBase

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(gHi gLo pHi pLo : ℝ) →  
  EconCSLib.Probability.weightedBernoulliFailureBase gHi gLo pHi pLo +  
  EconCSLib.Probability.weightedBernoulliSuccessBase gHi gLo pHi pLo
```

Exact Lean library declaration

```
def weightedBernoulliClosedRateBase (gHi gLo pHi pLo : ℝ) : ℝ :=  
  weightedBernoulliFailureBase gHi gLo pHi pLo +  
  weightedBernoulliSuccessBase gHi gLo pHi pLo
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.weightedBernoulliClosedThresholdRate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(gHi gLo pHi pLo : ℝ) → -(gHi + gLo) * Real.log (EconCSLib.Probability.weightedBernoulliClosedRateBase gHi gLo pHi pLo)
```

Exact Lean library declaration

```
def weightedBernoulliClosedThresholdRate (gHi gLo pHi pLo : ℝ) : ℝ :=  
  -(gHi + gLo) *  
    Real.log (weightedBernoulliClosedRateBase gHi gLo pHi pLo)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finiteMax

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} → [inst : Fintype α] → [inst_1 : Nonempty α] → (f : α → ℝ) → Finset.univ.sup' ... f
```

Exact Lean library declaration

```
def finiteMax {α : Type*} [Fintype α] [Nonempty α]  
  (f : α → ℝ) : ℝ :=  
  (Finset.univ : Finset α).sup' Finset.univ_nonempty f
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finiteMin

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} → [inst : Fintype α] → [inst_1 : Nonempty α] → (f : α → ℝ) → Finset.univ.inf' ... f
```

Exact Lean library declaration

```
def finiteMin {α : Type*} [Fintype α] [Nonempty α]  
  (f : α → ℝ) : ℝ :=  
  (Finset.univ : Finset α).inf' Finset.univ_nonempty f
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.strictUpperPairSet

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall (a : \mathbb{R} \times \mathbb{R}), \{q \mid q.2 < q.1\} a$

Exact Lean library declaration

```
def strictUpperPairSet : Set (ℝ × ℝ) :=  
  {q : ℝ × ℝ | q.2 < q.1}
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 187-204

Verbatim source input

3

Model and optimization

We now formalize our model and show how to optimize the rating function to maximize the learning rate. We focus on finding an optimal $\beta : [0, 1] \rightarrow [0, 1]$, a map from item quality θ to the probability it should receive a positive rating. This section requires no data: we characterize the optimal system.

3.1

Model and problem specification

Our model is constructed to emphasize the rating system's learning rate. Time is discrete ($k = 0, 1, 2, \dots$). Informally: there is a set of items. Each time step, buyers match with the items and leave a rating according to $\beta(\theta)$. The platform records the ratings and ranks the items. Formally:

Expanded PaperInterface specification

`GJ190ptimalBinaryRatingSystems.ProofBridge.source_quality_domain = GJ190ptimalBinaryRatingSystems.sourceQualityDomain`

Lean proof endpoint (built separately)

`GJ190ptimalBinaryRatingSystems.PaperInterface.source_quality_domain`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 199-218

Verbatim source input

Our model is constructed to emphasize the rating system's learning rate. Time is discrete ($k = 0, 1, 2, \dots$). Informally: there is a set of items. Each time step, buyers match with the items and leave a rating according to $\beta(\theta)$. The platform records the ratings and ranks the items. Formally:

existence of a nondecreasing match function $g(\theta)$, where item θ receives $n_k(\theta) = bkg(\theta)c$ matches, and thus ratings, up to time k . In other words, item θ is matched approximately every $g(\theta)$ time steps. $g(\theta) \leq 1$ and bounded away from 0, i.e. $\exists c > 0: g(\theta) > c$. This accumulation captures the feature that better items may be more likely to match. Ratings. The key quantity for our subsequent analysis is the probability of a positive rating for each θ , $\beta(\theta)$, $\Pr(\text{positive rating}|\theta)$. Let $y_k(\theta) \sim \text{Bernoulli}(\beta(\theta))$ be the rating an item of quality θ receives at the k th time it matches.

Expanded PaperInterface specification

```
GJ190ptimalBinaryRatingSystems.ProofBridge.source_matching_count = GJ190ptimalBinaryRatingSystems.sourceMatchingCount
```

Lean proof endpoint (built separately)

```
GJ190ptimalBinaryRatingSystems.PaperInterface.source_matching_count
```

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 199-218

Verbatim source input

Our model is constructed to emphasize the rating system's learning rate. Time is discrete ($k = 0, 1, 2, \dots$). Informally: there is a set of items. Each time step, buyers match with the items and leave a rating according to $\beta(\theta)$. The platform records the ratings and ranks the items. Formally:

existence of a nondecreasing match function $g(\theta)$, where item θ receives $n_k(\theta) = bkg(\theta)c$ matches, and thus ratings, up to time k . In other words, item θ is matched approximately every $g(\theta)$ time steps. $g(\theta) \leq 1$ and bounded away from 0, i.e. $\exists c > 0: g(\theta) > c$. This accumulation captures the feature that better items may be more likely to match. Ratings. The key quantity for our subsequent analysis is the probability of a positive rating for each θ , $\beta(\theta)$, $\Pr(\text{positive rating}|\theta)$. Let $y_k(\theta) \sim \text{Bernoulli}(\beta(\theta))$ be the rating an item of quality θ receives at the k th time it matches.

Expanded PaperInterface specification

```
GJ190ptimalBinaryRatingSystems.ProofBridge.source_matching_function =  
  GJ190ptimalBinaryRatingSystems.SourceMatchingFunction
```

Lean proof endpoint (built separately)

```
GJ190ptimalBinaryRatingSystems.PaperInterface.source_matching_function
```

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 219-237

Verbatim source input

Aggregating ratings and ranking sellers.
These ratings are aggregated into a reputation score,
 $x_k(\theta)$, at each time k . The score is the fraction
of positive ratings received up to time k : $x_k(\theta) = \frac{p_{nk}(\theta)}{n_k(\theta)}$
1
 $\hat{y}(\theta)$ with $x_0(\theta) = 0$ for all θ . Thus,
 $n_k(\theta) \sim \text{Binomial}(\beta(\theta), n_k(\theta))$.
 $x_k(\theta) \sim \text{Binomial}(\beta(\theta), n_k(\theta))$.
System state. The state of the system is given by a
joint distribution $\mu_k(\theta, X)$, which gives the mass of
items of quality $\theta \in \Theta \subset [0, 1]$ with aggregate score
 $x_k(\theta) \in X \subset [0, 1]$ at time k . Because our model is
a continuum, the evolution of the system state μ_k
follows a deterministic dynamical system.
We have described these dynamics at the level of individual items; however, such statements should be
interpreted as describing the evolution of the joint
distribution μ_k . The state update for μ_k is determined by the mass of items that match and the distributions of their ratings. A
 $\hookrightarrow$ formal description of
the state evolution is in Appendix Section B.1.

Expanded PaperInterface specification

```
GJ190ptimalBinaryRatingSystems.ProofBridge.source_empirical_reputation_score =  
  GJ190ptimalBinaryRatingSystems.sourceEmpiricalReputationScore
```

Lean proof endpoint (built separately)

```
GJ190ptimalBinaryRatingSystems.PaperInterface.source_empirical_reputation_score
```

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 228-237

Verbatim source input

System state. The state of the system is given by a joint distribution $\mu_k(\theta, X)$, which gives the mass of items of quality $\theta \in \Theta \subset [0, 1]$ with aggregate score $x_k(\theta) \in X \subset [0, 1]$ at time k . Because our model is a continuum, the evolution of the system state μ_k follows a deterministic dynamical system. We have described these dynamics at the level of individual items; however, such statements should be interpreted as describing the evolution of the joint distribution μ_k . The state update for μ_k is determined by the mass of items that match and the distributions of their ratings. A $\hookrightarrow$ formal description of the state evolution is in Appendix Section B.1.

Expanded PaperInterface specification

GJ190ptimalBinaryRatingSystems.ProofBridge.source_system_state = GJ190ptimalBinaryRatingSystems.SourceSystemState

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_system_state

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 238-256

Verbatim source input

Platform objective. The platform wishes to rank the items accurately. Given β and $\theta_1 > \theta_2$, define:
[U+0001 control character]
 $P_k(\theta_1, \theta_2 | \beta) = \mu_k x_k(\theta_1) > x_k(\theta_2) | \theta_1, \theta_2$
[U+0001 control character]
 $-\mu_k x_k(\theta_1) < x_k(\theta_2) | \theta_1, \theta_2$
(1)

Items. The system consists of a set $[0, 1]$ of items, where each item is associated with a unique (but unknown) quality $\theta \in [0, 1]$; i.e., the system consists of a continuum of a unit mass of items whose unknown qualities are uniform² in $[0, 1]$. Below, we discretize the continuous quality space $[0, 1]$ into M types, to calculate a stepwise increasing β . We will make clear why we introduce a continuum but then discretize.

This expression captures observed score ranking's accuracy. When $\theta_1 > \theta_2$ but $x_k(\theta_1) < x_k(\theta_2)$, the ranking mistakenly orders θ_1 below θ_2 . A good system has large $P_k(\theta_1, \theta_2 | \beta)$. Integrating across items

Expanded PaperInterface specification

$\forall$ (probabilityCorrect probabilityIncorrect : $\mathbb{R}$),
GJ190ptimalBinaryRatingSystems.ProofBridge.source_definition_pairwise_accuracy_eq1 probabilityCorrect
probabilityIncorrect =
probabilityCorrect - probabilityIncorrect

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_definition_pairwise_accuracy_eq1

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 257-288

Verbatim source input

creates the following objective for each time k :
$$W_k = \sum_{\theta_1, \theta_2} w(\theta_1, \theta_2) P_k(\theta_1, \theta_2 | \beta) d_{\theta_1} d_{\theta_2}$$

(2)

Matching with buyers. Items accumulate ratings over time by matching with buyers. We assume the

Weight function $w(\theta_1, \theta_2) > 0$ indicates how much the platform cares about not mistaking a quality θ_1 item with a quality θ_2 item. We consider scaled w such that $\theta_1 > \theta_2$ $w(\theta_1, \theta_2) d_{\theta_1} d_{\theta_2} = 1$.

1
Note that a rating is not the same as a reward; buyers often give positive ratings after bad experiences.
2
Any distribution can be handled by considering θ to be the item's quantile rather than its absolute quality.
 $\theta_1 > \theta_2$

Our first question then becomes: What β yields the highest value of W_k ? As discussed above, the plat-

[form-feed in source extraction]
Designing Optimal Binary Rating Systems

form influences β through the design of its rating system. The optimal choice of β sets the benchmark. Discussion. Objective function. The specification (2) of the objective is quite rich. It contains scaled

Expanded PaperInterface specification

$\forall$ (weight pairwiseAccuracy : $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$) (Wk : $\mathbb{R}$),
(GJ190OptimalBinaryRatingSystems.ProofBridge.source_definition_weighted_objective_eq2 weight pairwiseAccuracy Wk =
 $\forall \theta_1 \in \text{Set.Icc } 0 \ 1, \forall \theta_2 \in \text{Set.Ico } 0 \ \theta_1, 0 < \text{weight } \theta_1 \ \theta_2 \wedge$
 $\int (\theta_1 : \mathbb{R}) \text{ in Set.Icc } 0 \ 1, \int (\theta_2 : \mathbb{R}) \text{ in Set.Ico } 0 \ \theta_1, \text{weight } \theta_1 \ \theta_2 = 1 \wedge$
 $W_k = \int (\theta_1 : \mathbb{R}) \text{ in Set.Icc } 0 \ 1, \int (\theta_2 : \mathbb{R}) \text{ in Set.Ico } 0 \ \theta_1, \text{weight } \theta_1 \ \theta_2 * \text{pairwiseAccuracy } \theta_1 \ \theta_2$

Lean proof endpoint (built separately)

GJ190OptimalBinaryRatingSystems.PaperInterface.source_definition_weighted_objective_eq2

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 321-360

Verbatim source input

Large deviations & discretization

Recall the question: What β yields the highest value of W_k ? We now refine objective W_k and constrain β to form a non-degenerate, feasible optimization task. Large deviation rate function. W_k is not one objective: it has a different value per time k , and no single β simultaneously optimizes W_k for all

3
We use θ_1, θ_2 ($\theta_1 - \theta_2$), $(1 - \theta_1)(1 - \theta_2)(\theta_1 - \theta_2)$, and 1
 $(2 - \theta_1)^2(1 - \theta_2)^2(\theta_1 - \theta_2)$ as examples.

k.4 Considering asymptotic performance is also insufficient: when β is strictly increasing in θ , $\lim_{k \rightarrow \infty} P_k(\theta_1, \theta_2 | \beta) = 1 \forall \theta_1, \theta_2$ by the law of large numbers. Thus, W , $\lim_{k \rightarrow \infty} W_k = 1$, and any such β is asymptotically optimal.

For this reason, we consider maximization of the rate at which W_k converges, i.e., how fast the estimated item ranking converges to the true item ranking. We use a large

deviations approach (Dembo and Zeitouni, 2010) to quantify this convergence rate. Formally, given sequence $Y_k \leq \lim_{k \rightarrow \infty} Y_k = Y$, the large deviations rate of convergence is $-\lim_{k \rightarrow \infty} k \log(Y - Y_k) = c$. If c exists then Y_k approaches Y exponentially fast: $Y - Y_k = e^{-kc+o(k)}$.

Then, we wish to choose β to maximize W_k 's large deviations rate, $r = -\lim_{k \rightarrow \infty} k \log(W - W_k)$. Discretizing β . Unfortunately, even this problem is degenerate if we consider continuous β : for any β that is not piecewise constant, the large deviations rate of convergence is zero, i.e., convergence of W_k to its limit is only polynomially fast, and characterizing the dependence of this convergence rate on β is intractable. Thus, the rate of convergence for W_k is not a satisfactory objective with continuous β .

We make progress by discretizing β ; in particular, we restrict attention to optimization over stepwise increasing β functions. Among stepwise increasing β , the large deviations rate of W_k to its limiting value W can be shown to be nondegenerate, i.e. $\exists c > 0$ s.t. $W - W_k = e^{-kc+o(k)}$. (See Lemma C.4 in the Appendix for further discussion.)

Expanded PaperInterface specification

```
∀ (W : ℝ) (Wk : ℕ → ℝ) (rate : ℝ),
  GJ190ptimalBinaryRatingSystems.ProofBridge.source_definition_large_deviation_rate W Wk rate =
    EconCSLib.Probability.HasExponentialRate (fun k => W - Wk k) rate
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_definition_large_deviation_rate

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 354-368

Verbatim source input

We make progress by discretizing β ; in particular, we restrict attention to optimization over stepwise increasing β functions.⁵ Among stepwise increasing β , the large deviations rate of W_k to its limiting value W can be shown to be nondegenerate, i.e. $\exists c > 0$ s.t. $W - W_k = e^{-kc+o(k)}$. (See Lemma C.4 in the Appendix for further discussion.)

Notationally, we will calculate an optimal stepwise increasing β with M levels, i.e. there are M intervals $S_i \subset [0, 1]$ and levels t_i such that when $\theta_1, \theta_2 \in S_i$, then $\beta(\theta_1) = \beta(\theta_2) = t_i$. The challenge is calculating an optimal $S^* = \{S_i\}$ and $t^* = \{t_i\}$. The physical interpretation is that we group the items into M subsets (types) $S_i \subset [0, 1]$. When items θ_1, θ_2 are in the same subset, then their asymptotic reputation scores are the same, $\lim_k x_k(\theta_1) = \lim_k x_k(\theta_2) = t_i$. These items cannot be distinguished from one another even asymptotically.

Expanded PaperInterface specification

$\forall (m : \mathbb{N}) (\text{cutpoints} : \mathbb{N} \rightarrow \mathbb{R}) (\text{levels} : \text{Fin } (m + 2) \rightarrow \mathbb{R}),$
 $\text{GJ190ptimalBinaryRatingSystems.ProofBridge.source_definition_step_rule_partition_levels } m \text{ cutpoints levels} =$
 $\text{GJ190ptimalBinaryRatingSystems.intervalCutpointsEndpointFeasible } m \text{ cutpoints } \wedge$
 $\text{GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector levels}$

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_definition_step_rule_partition_levels

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 459-476

Verbatim source input

Our optimization problem: Within the class of stepwise increasing functions with M levels, find the β that is optimal, i.e. is

For Kendall's τ and Spearman's ρ , the optimal intervals are simply equispaced in $[0, 1]$, i.e. $S_i^* = [M_i, i+1)$, because the entire item quality distribution is equally important. For other objective weight functions w , the difficulty of finding the optimal subsets S^* depends on the properties of w . Since S^* is trivial for Kendall's τ and Spearman's ρ – and w is just an analytic tool that formalizes a platform's goals – we focus on finding the optimal levels t^* .

(1) Asymptotically optimal. It yields the highest limiting value of W_k . AND
(2) Rate optimal. It yields the fastest large deviations rate r among asymptotically optimal β .
[U+0001 control character]
A remarkable result of our paper is a $O(M \log^2 M)$
[U+000F control character]
procedure to find an optimal β with M levels.

Expanded PaperInterface specification

$\forall \{\alpha : \text{Type } u_1\} \text{ (feasible : } \alpha \rightarrow \text{Prop) (primary secondary : } \alpha \rightarrow \mathbb{R}) (x : \alpha),$
feasible $x \wedge \forall (y : \alpha), \text{ feasible } y \rightarrow \text{primary } y < \text{primary } x \vee \text{primary } y = \text{primary } x \wedge \text{secondary } y \leq \text{secondary } x$

Lean proof endpoint (built separately)

GJ190OptimalBinaryRatingSystems.PaperInterface.source_definition_lexicographic_optimality

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 488-507

Verbatim source input

Theorem 3.1. The β defined by the following
Suppose we started with a set of M items. Then the
choices of S^* and t^* is optimal:
only remaining challenge is to equalize the rates at
 R
 P
which each item is separated from others: the large
 $*$
 $S = \arg \max_{0 \leq i < j < M} \theta_2 \in S_i, \theta_1 \in S_j w(\theta_1, \theta_2) d(\theta_1, \theta_2)$
deviations rate is unaffected by the weight function
 $*$
 6
 w (it does not appear in the simplification of $r(t)$).
 $t = \arg \max r(t)$, where $g_i, \inf_{\theta \in S_i} g(\theta)$ and
In other words, given a discrete set of M items
 1
(equiv, given S^*), calculating the optimal t is equivr(t), $-\lim_{k \rightarrow \infty} \log(W - W_k)$
(3)
alent to solving a maximin problem for the rates at
 $k \rightarrow \infty$ k

Expanded PaperInterface specification

```
∀ (μ : MeasureTheory.Measure ℝ) [MeasureTheory.IsFiniteMeasure (μ.prod μ)] [(μ.prod μ).IsOpenPosMeasure]
(S : GJ190ptimalBinaryRatingSystems.Theorem31SourceFiniteDiscretizationWeightedModel μ) (limitingValue : (ℕ → ℝ) → ℝ)
(rate : (ℕ → ℝ) → (Fin (S.m + 2) → ℝ) → ℝ),
EconCSLib.Optimization.IsMaximizerOn
(GJ190ptimalBinaryRatingSystems.monotoneIntervalCutpointsEndpointFeasible (S.m + 2)) limitingValue S.cut →
(∀ (cut : ℕ → ℝ),
  GJ190ptimalBinaryRatingSystems.monotoneIntervalCutpointsEndpointFeasible (S.m + 2) cut →
    limitingValue cut = limitingValue S.cut → cut = S.cut) →
(∀ (levels : Fin (S.m + 2) → ℝ),
  GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector levels →
    rate S.cut levels =
      GJ190ptimalBinaryRatingSystems.binaryEndpointAwareAdjacentRateObjective levels S.sampleRate) →
  ∃ levels,
  ∃ (hlevels : GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector levels),
  GJ190ptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize levels S.sampleRate ∧
  EconCSLib.Probability.ExponentialRateCertificate
    (GJ190ptimalBinaryRatingSystems.theorem31SourceWbar μ S.cut ~ S.sampleRate levels hlevels S.weight)
    (GJ190ptimalBinaryRatingSystems.binaryEndpointAwareAdjacentRateObjective levels S.sampleRate) ∧
  EconCSLib.Optimization.IsLexicographicMaximizerOn
    (fun design =>
      GJ190ptimalBinaryRatingSystems.monotoneIntervalCutpointsEndpointFeasible (S.m + 2) design.1 ∧
      GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector design.2)
    (fun design => limitingValue design.1) (fun design => rate design.1 design.2) (S.cut, levels))
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.theorem31_source_matching_function_unique_value_argmax_lexicographic

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 501-518

Verbatim source input

$t = \arg \max r(t)$, where g_i , $\inf_{\theta \in S_i} g(\theta)$ and
In other words, given a discrete set of M items
1
(equiv, given S^*), calculating the optimal t is $\text{equivr}(t)$, $-\lim \log(W - W_k)$
(3)
alent to solving a maximin problem for the rates at
 $k \rightarrow \infty$ k
which
each type is distinguished from each of the
 $= \min \inf \{g_{i+1} KL(a||t_{i+1}) + g_i KL(a||t_i)\}$
 $0 \leq i \leq M-2$ $a \in R$
others. Thus, the algorithm below also solves the
inverse bandits problem in which we wish to rank
The proof is in the Appendix. The main hurdle is
the M arms, and we can choose the structure of the
showing that the continuum of rates for $P_k(\theta_1, \theta_2)$
(binary) observations at each arm.
for each pair θ_1, θ_2 translates into a rate for W_k .

Expanded PaperInterface specification

```
∀ {m : ℕ} (cutpoints : ℕ → ℝ) (g : ℝ → ℝ) (j : Fin (m + 1)),
  Monotone cutpoints →
  Monotone g →
  ∀ (tHi tLo : ℝ),
    GJ190ptimalBinaryRatingSystems.sourceCellMatchingRate cutpoints g
      (GJ190ptimalBinaryRatingSystems.adjacentLowIndex j) =
      g (cutpoints ↑ (GJ190ptimalBinaryRatingSystems.adjacentLowIndex j)) ∧
    GJ190ptimalBinaryRatingSystems.sourceCellMatchingRate cutpoints g
      (GJ190ptimalBinaryRatingSystems.adjacentHighIndex j) =
      g (cutpoints ↑ (GJ190ptimalBinaryRatingSystems.adjacentHighIndex j)) ∧
    GJ190ptimalBinaryRatingSystems.paperAdjacentBinaryRatingRate
      (GJ190ptimalBinaryRatingSystems.sourceCellMatchingRate cutpoints g
        (GJ190ptimalBinaryRatingSystems.adjacentHighIndex j))
      (GJ190ptimalBinaryRatingSystems.sourceCellMatchingRate cutpoints g
        (GJ190ptimalBinaryRatingSystems.adjacentLowIndex j))
      tHi tLo =
    sInf
      (Set.range fun a =>
        GJ190ptimalBinaryRatingSystems.sourceCellMatchingRate cutpoints g
          (GJ190ptimalBinaryRatingSystems.adjacentHighIndex j) *
          GJ190ptimalBinaryRatingSystems.paperBernoulliKL a tHi +
          GJ190ptimalBinaryRatingSystems.sourceCellMatchingRate cutpoints g
            (GJ190ptimalBinaryRatingSystems.adjacentLowIndex j) *
            GJ190ptimalBinaryRatingSystems.paperBernoulliKL a tLo)
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_theorem31_adjacent_rate_eq3

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 501-518

Verbatim source input

$t = \arg \max r(t)$, where g_i , $\inf_{\theta \in S_i} g(\theta)$ and
In other words, given a discrete set of M items
1
(equiv, given $S *$), calculating the optimal t is $\text{equivr}(t)$, $-\lim \log(W - W_k)$
(3)
alent to solving a maximin problem for the rates at
 $k \rightarrow \infty$ k
which
each type is distinguished from each of the
 $= \min \inf \{g_{i+1} KL(a||t_{i+1}) + g_i KL(a||t_i)\}$
 $0 \leq i \leq M-2$ $a \in R$
others. Thus, the algorithm below also solves the
inverse bandits problem in which we wish to rank
The proof is in the Appendix. The main hurdle is
the M arms, and we can choose the structure of the
showing that the continuum of rates for $P_k(\theta_1, \theta_2)$
(binary) observations at each arm.
for each pair θ_1, θ_2 translates into a rate for W_k .

Expanded PaperInterface specification

$\forall \{m : \mathbb{N}\} (cut : \mathbb{N} \rightarrow \mathbb{R}) (g : \mathbb{R} \rightarrow \mathbb{R}) (i : \text{Fin } (m + 2)),$
 $cut \uparrow i \leq cut (\uparrow i + 1) \rightarrow$
 $\text{MonotoneOn } g (\text{Set.Icc } (cut \uparrow i) (cut (\uparrow i + 1))) \rightarrow$
 $\text{GJ190ptimalBinaryRatingSystems.sourceCellMatchingRate } cut \ g \ i = g (cut \uparrow i)$

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.theorem31_source_cell_matching_rate_eq_lower_cutpoint

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 519-526

Verbatim source input

We further note that the choice of M is not consequential; in the Appendix Section B.4 we show that $KL(a||b) = a \log a + (1 - a) \log 1-a$ is the Kullback-Leibler (KL) divergence between Bernoulli random variables with success probabilities a and b respectively. In an appropriate sense, a sequence of optimal β_M converges as M gets large.

Expanded PaperInterface specification

```
∀ (a b : ℝ),
  GJ190ptimalBinaryRatingSystems.paperBernoulliKL a b = a * Real.log (a / b) + (1 - a) * Real.log ((1 - a) / (1 - b))
```

Lean proof endpoint (built separately)

```
GJ190ptimalBinaryRatingSystems.PaperInterface.definition_bernoulli_kl_formula
```

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 530-549

Verbatim source input

Algorithm to find the optimal levels We now describe how to find t^* , the maximizer of $r(t)$. The following lemma describes a system of equations to find the t^* that maximizes r . It states that t^* equalizes the rates at which each interval i is separated from its neighbors. The proof involves manipulation of r and convexity, $\hookrightarrow$ and is in the appendix.

Lemma 3.1. The unique solution t^* to the following system of equations maximizes $r(t)$:

$$r(t_0, t_1) = r(t_1, t_2) = \dots = r(t_{M-2}, t_{M-1})$$

(4)

$t_0 = 0, t_{M-1} = 1$

[U+0014 control character][U+0012 control character]

g_{i-1}

g_i

$r(t_{i-1}, t_i), -\log(1 - t_{i-1}) g_{i-1} + g_i(1 - t_i) g_{i-1} + g_i$

g_{i-1}

g_i

Expanded PaperInterface specification

```
∀ {m : ℕ},
  1 < m →
    ∀ (sampleRate : Fin (m + 2) → ℝ),
      (∀ k < m + 2, 0 < GJ190ptimalBinaryRatingSystems.binaryEndpointSampleRateNat sampleRate k) →
        ∃! levels,
          GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector levels ^
            GJ190ptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize levels sampleRate ^
              EconCSLib.Optimization.IsMaximizerOn GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector
                (fun xs => GJ190ptimalBinaryRatingSystems.binaryEndpointAwareAdjacentRateObjective xs sampleRate) levels
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_lemma31_equalization_eq4

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 542-571

Verbatim source input

```
[U+0014 control character][U+0012 control character]
gi-1
gi
r(ti-1 , ti ) , - log (1 - ti-1 ) gi-1 +gi (1 - ti ) gi-1 +gi
gi-1

gi

ALGORITHM 1: Nested Bisection
Data: Number of intervals M , match function g
Result:  $\beta$  levels, i.e.  $\{t_0 \dots t_M - 1\}$ 
Function main (M ,  $\delta$ , g)
while Uncertainty region for  $t_M - 2$  is bigger
than error tolerance do
  Calculate  $r(t_M - 2 , 1)$ , the rate between
  current guess for  $t_M - 2$  , and  $t_M - 1 = 1$ .
  Fixing  $t_M - 2$  , find  $t_1 \dots t_M - 3$  such that
   $r(t_1 , t_2 ) = r(t_2 , t_3 ) = \dots =$ 
   $r(t_M - 3 , t_M - 2 ) = r(t_M - 2 , 1)$ , which can be
  done through a sequence of bisection
  routines.
  Calculate  $r(0 , t_1 )$ , the rate between current
  guess for  $t_1$  , and  $t_0 = 0$ .
  Compare  $r(t_M - 2 , 1)$  and  $r(0 , t_1 )$ , adjust
  uncertainty region for  $t_M - 2$  accordingly.
  return  $\{t_i\}$ 

[U+0013 control character]gi-1 +gi [U+0015 control character]

+ti-1 gi-1 +gi ti gi-1 +gi
```

Expanded PaperInterface specification

```
∀ (gLo gHi tLo tHi : ℝ),
  GJ190OptimalBinaryRatingSystems.paperAdjacentBinaryClosedRate gLo gHi tLo tHi =
    -(gLo + gHi) *
      Real.log
        ((1 - tLo) ^ (gLo / (gLo + gHi)) * (1 - tHi) ^ (gHi / (gLo + gHi)) +
          tLo ^ (gLo / (gLo + gHi)) * tHi ^ (gHi / (gLo + gHi)))
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.lemma31_closed_adjacent_rate_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 550-587

Verbatim source input

ALGORITHM 1: Nested Bisection
Data: Number of intervals M , match function g
Result: β levels, i.e. $\{t_0 \dots t_M - 1\}$
Function main (M, δ, g)
while Uncertainty region for $t_M - 2$ is bigger
than error tolerance do
 Calculate $r(t_M - 2, 1)$, the rate between
 current guess for $t_M - 2$, and $t_M - 1 = 1$.
 Fixing $t_M - 2$, find $t_1 \dots t_{M-3}$ such that
 $r(t_1, t_2) = r(t_2, t_3) = \dots =$
 $r(t_{M-3}, t_{M-2}) = r(t_{M-2}, 1)$, which can be
 done through a sequence of bisection
 routines.
 Calculate $r(0, t_1)$, the rate between current
 guess for t_1 , and $t_0 = 0$.
 Compare $r(t_M - 2, 1)$ and $r(0, t_1)$, adjust
 uncertainty region for $t_M - 2$ accordingly.
return $\{t_i\}$

[U+0013 control character]gi-1 +gi [U+0015 control character]

+ti-1 gi-1 +gi ti gi-1 +gi
We do not know of any algorithm that efficiently
and provably solves such convex equality systems
in general. However, we leverage some structure in
our setting to develop an algorithm, NestedBisection, with run-time and optimality guarantees. The
efficiency of our algorithm results from the property
that, given a rate, t_i is uniquely determined by the
value of either of the adjacent levels t_{i-1} , t_{i+1} , reducing an exponentially large search space to an almost
linear one. Physically, i.e., we only need to separate
each type of item from its neighboring types.
Below we include pseudo-code. Akin to branch and
bound, the algorithm proceeds via bisection on the
optimal value of $t_M - 2$. For each candidate value
of $t_M - 2$, the other values can be found using a sequence of bisection subroutines. These values approximately obey all the
↪ equalities in the system (4)
except the first. The direction of the first equality's
violation reveals how to change the interval for the
next outer bisection iteration.

Expanded PaperInterface specification

```
∀ (m outerSteps innerSteps : ℕ) (sampleRate : Fin (m + 2) → ℝ) (grid : ℝ),
  GJ190ptimalBinaryRatingSystems.theorem32WeightedNestedBisectionOutput m outerSteps innerSteps sampleRate grid =
    have gLast :=
      sampleRate (GJ190ptimalBinaryRatingSystems.adjacentLowIndex GJ190ptimalBinaryRatingSystems.lastAdjacentIndex);
    have candidate := fun lastLow =>
      GJ190ptimalBinaryRatingSystems.theorem32WeightedBackwardGridRateBisectionLevels m innerSteps sampleRate grid
        (gLast * -Real.log lastLow) lastLow;
    have above := fun lastLow =>
      GJ190ptimalBinaryRatingSystems.theorem32OuterSourceWeightedRateAbove (candidate lastLow) sampleRate;
    have lastLow := (EconCSLib.Optimization.realBisectionRun above outerSteps (1 - 1 / ↑(m + 1)) (1 - grid)).2;
    candidate lastLow
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.theorem32_weighted_nested_bisection_output

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 18

Source locator: cited publication, lines 588-598

Verbatim source input

Theorem 3.2. [U+0001 control character]NestedBisection finds an [U+000F control character]-optimal t in $O(M \log^2 M)$ operations, where [U+000F control character]-optimal means [U+000F control character] that $r(t)$ is within additive constant [U+000F control character] of optimal. The proof is in the appendix. The main difficulty is finding a Lipschitz constant [U+000F control character](δ) for how much the rate changes with a shift δ in a level t_i . This requires lower bounding t_1 as a function of M . In practice, the algorithm runs instantaneously on a modern machine (e.g. for $M = 200$).

Expanded PaperInterface specification

```

∀ {m : ℕ},
  1 < m →
    ∀ (optimal sampleRate : Fin (m + 2) → ℝ) (eps : ℝ),
      GJ190OptimalBinaryRatingSystems.BinaryEndpointLevelVector optimal →
        GJ190OptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize optimal sampleRate →
          (∀ (idx : Fin (m + 2)), 0 < sampleRate idx) →
            (∀ {a b : Fin (m + 2)}, ↑a ≤ ↑b → sampleRate a ≤ sampleRate b) →
              sampleRate
                (GJ190OptimalBinaryRatingSystems.adjacentHighIndex
                  GJ190OptimalBinaryRatingSystems.firstAdjacentIndex) =
                  1 →
                    0 < eps →
                      have grid := GJ190OptimalBinaryRatingSystems.theorem32WeightedSourceGrid sampleRate eps;
                      have lastLower := 1 - 1 / ↑(m + 1);
                      have sourceDepthBudget := max ((1 - grid - lastLower) / 2) 1;
                      have runtimeLog := Real.logb 2 (max 1 (sourceDepthBudget / grid)) + 2;
                      0 < grid ∧
                        ∃ L,
                          ↑(L + 1) ≤ runtimeLog ∧
                            GJ190OptimalBinaryRatingSystems.binaryEndpointAwareAdjacentRateObjective optimal sampleRate -
                              GJ190OptimalBinaryRatingSystems.binaryEndpointAwareAdjacentRateObjective
                                (GJ190OptimalBinaryRatingSystems.theorem32WeightedNestedBisectionOutput m (L + 1) L
                                  sampleRate grid)
                                  sampleRate ≤
                                    eps ∧
                                      GJ190OptimalBinaryRatingSystems.nestedBisectionOperationCount (m + 2) (L + 1) L ≤
                                        EconCSLib.Optimization.nestedBisectionStepBound (m + 2) L ∧
                                          ↑(GJ190OptimalBinaryRatingSystems.nestedBisectionOperationCount (m + 2) (L + 1) L) ≤
                                            ↑(m + 2) * runtimeLog ^ 2

```

Lean proof endpoint (built separately)

GJ190OptimalBinaryRatingSystems.PaperInterface.theorem32_weighted_nested_bisection_loss_and_runtime

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 642-668

Verbatim source input

Resources available to platform. We suppose the platform has a set Y of possible binary questions that it could ask a buyer, e.g., “Are you satisfied with your experience” $\hookrightarrow$ or “Is this experience your best on our platform?”. Informally, at each rating opportunity (i.e., match made), the rater can be shown a single question $y \in Y$. Let $\psi(\theta, y)$ be the probability an item quality θ would receive a positive rating when the rater is asked question $y \in Y$. We further suppose the platform has a set Θ of representative items for which it has access to item quality. θ , then, is the $\hookrightarrow$ granularity at which the platform can collect data about historical performance, or otherwise get expert labels. (We assume M [U+001D control character] $|\Theta|$). Using known set Θ , the platform can run an experiment to create an estimate $\hat{\psi}(\theta, y)$, $\forall y \in Y, \theta \in \Theta$. Design heuristic. How can the platform build an effective rating system using these primitives and $\hat{\beta}$? We consider the following heuristic design: the platform randomly shows a question $y \in Y$ to each buyer. The choice of the platform is a distribution $H(y)$ over $y \in Y$; in other words, for the platform the design of the system amounts to choosing the frequency with which each question is shown. At each rating opportunity, y is chosen from Y according to H , independently across opportunities. Clearly, the probability that P an item θ receives a positive rating is: $\beta(\theta) = \sum_{y \in Y} \psi(\theta, y)H(y)$. We want a distribution H such that $\hat{\beta}(\theta) = \beta(\theta)$ for all $\theta \in [\theta, 1]$, i.e., that the positive rating probability for each item is exactly the optimal value. However, there may not exist any set of questions Y with associated ψ and choice of H such that $\hat{\beta} = \beta$.7

Expanded PaperInterface specification

$\forall \{Y : \text{Type } u_1\} \text{ [inst : Fintype } Y] (H : Y \rightarrow \mathbb{R}), (\forall (y : Y), 0 \leq H y) \wedge \sum y, H y = 1$

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.section4_question_distribution

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 20

Source locator: cited publication, lines 652-668

Verbatim source input

Using known set θ , the platform can run an experiment to create an estimate $\psi(\theta, y)$, $\forall y \in Y, \theta \in \Theta$. Design heuristic. How can the platform build an effective rating system using these primitives and β ? We consider the following heuristic design: the platform randomly shows a question $y \in Y$ to each buyer. The choice of the platform is a distribution $H(y)$ over $y \in Y$; in other words, for the platform the design of the system amounts to choosing the frequency with which each question is shown. At each rating opportunity, y is chosen from Y according to H , independently across opportunities. Clearly, the probability that θ receives a positive rating is: $\beta(\theta) = \sum_{y \in Y} \psi(\theta, y)H(y)$. We want a distribution H such that $\beta(\theta) = \beta^*(\theta)$ for all $\theta \in [0, 1]$, i.e., that the positive rating probability for each item is exactly the optimal value. However, there may not exist any set of questions Y with associated ψ and choice of H such that $\beta^* = \beta$.

Expanded PaperInterface specification

$\forall \{Y : \text{Type } u\ 1\} \text{ [inst : Fintype } Y] (\psi : \mathbb{R} \rightarrow Y \rightarrow \mathbb{R}) (H : Y \rightarrow \mathbb{R}) (\theta : \mathbb{R}),$
 $\text{GJ190ptimalBinaryRatingSystems.sourceInducedBinaryResponse } \psi \ H \ \theta = \sum y, \psi \ \theta \ y * H \ y$

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.section4_induced_binary_response

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 21

Source locator: cited publication, lines 669-696

Verbatim source input

We propose the following heuristic to address this difficulty. Choose a probability distribution H to minimize the following L1 distance:

$$\begin{aligned} & \sum_{\theta \in \Theta} \min_{y \in Y} |\beta(\theta) - \psi(\theta, y)H(y)| \\ & \text{(5)} \\ & H: \Theta \times Y \rightarrow \mathbb{R}^+ \end{aligned}$$

This heuristic uses the data available to the platform, $\psi(\theta, y)$ for a set of items $\theta \in \Theta$, and designs H so at least these items receive ratings close to their optimal ratings $\beta(\theta)$. Then, as long as ψ is wellbehaved, and Θ is representative of the full set, one can hope that $\beta(\theta) \approx \beta(\theta)$, for all $\theta \in [\Theta, 1]$. Discussion. Real-world analogue & constraints. A special case of this system is already in place on

There are special cases where an exact solution exists. For example, let $Y = [0, 1]$, and $\psi(\theta, y) = I[\theta \geq y]$. many platforms, where the same question is always shown. Static systems can be designed by restricting H to only have mass at one question y . More generally, constraints can be used in optimization (5).

Expanded PaperInterface specification

$\forall \{ \text{Representative} : \text{Type } u_1 \} \{ Y : \text{Type } u_2 \} [\text{inst} : \text{Fintype Representative}] [\text{inst}_1 : \text{Fintype } Y]$
 (quality : Representative $\rightarrow \mathbb{R}$) ($\beta : \mathbb{R} \rightarrow \mathbb{R}$) ($\psi_{\text{Hat}} : \text{Representative} \rightarrow Y \rightarrow \mathbb{R}$) ($H : Y \rightarrow \mathbb{R}$),
 GJ190ptimalBinaryRatingSystems.sourceQuestionDesignL1Objective quality β ψ_{Hat} $H =$
 $\sum_i i, |\beta(\text{quality } i) - \sum_y \psi_{\text{Hat}} i y * H y|$

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.section4_l1_design_objective

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 22

Source locator: cited publication, lines 669-696

Verbatim source input

We propose the following heuristic to address this difficulty. Choose a probability distribution H to minimize the following L1 distance:

$$\begin{aligned} & \sum_{\theta \in \Theta} \sum_{y \in Y} |\beta(\theta) - \psi(\theta, y)H(y)| \\ & (5) \\ & H: \Theta \times Y \rightarrow \mathbb{R}^+ \end{aligned}$$

This heuristic uses the data available to the platform, $\psi(\theta, y)$ for a set of items $\theta \in \Theta$, and designs H so at least these items receive ratings close to their optimal ratings $\beta(\theta)$. Then, as long as ψ is wellbehaved, and Θ is representative of the full set, one can hope that $\beta(\theta) \approx \beta(\theta)$, for all $\theta \in [\Theta, 1]$. Discussion. Real-world analogue & constraints. A special case of this system is already in place on

There are special cases where an exact solution exists. For example, let $Y = [0, 1]$, and $\psi(\theta, y) = I[\theta \geq y]$.

many platforms, where the same question is always shown. Static systems can be designed by restricting H to only have mass at one question y . More generally, constraints can be used in optimization (5).

Expanded PaperInterface specification

```
∀ {Representative : Type u_1} {Y : Type u_2} [inst : Fintype Representative] [inst_1 : Fintype Y]
  (quality : Representative → ℝ) (β : ℝ → ℝ) (ψHat : Representative → Y → ℝ) (H : Y → ℝ),
  EconCSLib.Optimization.IsMinimizerOn GJ190ptimalBinaryRatingSystems.SourceQuestionDistribution
    (GJ190ptimalBinaryRatingSystems.sourceQuestionDesignL1Objective quality β ψHat) H
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.section4_question_design_solution

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 23

Source locator: cited publication, lines 1820-1828

Verbatim source input

Formal specification of system state update

Recall that $\mu_k(\theta, X)$ is the mass of items with true quality $\theta \in \Theta \subseteq [0, 1]$ and a reputation score $x \in X \subseteq [0, 1]$ at time k . Let $E_k = \{\theta : n_k(\theta) = n_{k-1}(\theta) + 1\}$. These are the items who receive an additional rating at time k ; for all $\theta \in E_k$, $n_k(\theta) = n_{k-1}(\theta)$. Our system is completely deterministic, and evolves according to the distributions of the individual seller dynamics. For each $\theta \in E_k$, x, x_0 , define $\omega(\theta, x, x_0)$ as follows:
 $\omega(\theta, x, x_0) = \beta(\theta)I\{n_k(\theta)x - n_{k-1}(\theta)x_0 = 1\} + (1 - \beta(\theta))I\{n_k(\theta)x - n_{k-1}(\theta)x_0 = 0\}.$

Expanded PaperInterface specification

GJ190ptimalBinaryRatingSystems.ProofBridge.appendixB1_active_set =
GJ190ptimalBinaryRatingSystems.sourceAppendixBActiveSet

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.appendixB1_active_set

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 24

Source locator: cited publication, lines 1828-1831

Verbatim source input

$\omega(\theta, x, x_0) = \beta(\theta)I\{n_k(\theta)x - n_{k-1}(\theta)x_0 = 1\} + (1 - \beta(\theta))I\{n_k(\theta)x - n_{k-1}(\theta)x_0 = 0\}$.
Then ω gives the probability of transition from x_0 to x when an item receives a rating. We then have:
 $\sum_Z$

Expanded PaperInterface specification

GJ190ptimalBinaryRatingSystems.ProofBridge.appendixB1_transition_kernel =
GJ190ptimalBinaryRatingSystems.sourceAppendixBTransitionKernel

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.appendixB1_transition_kernel

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 25

Source locator: cited publication, lines 1832-1852

Verbatim source input

$$\begin{aligned} & \sum_{k=1}^Z \sum_{x \in X} \mu_k(\theta, x) \omega(\theta, x, x_0) dx \mu_k(dx_0, d\theta) + \\ & \mu_k(dx, d\theta). \end{aligned}$$

It is straightforward but tedious to check that the preceding dynamics are well defined, given our primitives.

Expanded PaperInterface specification

GJ190OptimalBinaryRatingSystems.ProofBridge.appendixB1_state_update =
GJ190OptimalBinaryRatingSystems.sourceAppendixBStateUpdateMass

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.appendixB1_state_update

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 26

Source locator: cited publication, lines 1927-1949

Verbatim source input

Formalization of effect of matching rates shifting

Matching concentrating at the top items moves mass of $\beta(\theta)$ away from high θ , and subsequently mass of $H(y)$ away from the questions that help distinguish the top items, as observed in Figures 1b and 3b above. Informally, $\hookrightarrow$ this occurs because when matching concentrates, top items are accumulating many ratings more ratings comparatively, and so the amount of information needed per rating is comparatively less. We formalize this intuition in Lemma B.1 below. The lemma states that if matching rates shift such that there is an index k above which matching rates increase and below which they decrease, then correspondingly the levels of β , (i.e. t_i) become closer together above k . Lemma B.1. Suppose $k, g, \tilde{g}$ such that $\forall j \in \{k+1 \dots M-1\}, g_j \geq \tilde{g}_j$, and $\forall j \in \{0 \dots k-1\}, g_j \leq \tilde{g}_j$, and $g_k = \tilde{g}_k$. Then, $t^*_k \geq \tilde{t}^*_k$. Proof. This proof is similar to that of Lemma 3.1, except that with the matching function changing the overall rate function can either increase, decrease, or stay the same. Suppose the overall rate function decreased or stayed the same when the matching function changed from $\tilde{g}$ to g . Then $g_{M-2} > \tilde{g}_{M-2}$ and the target rate is no larger, and so $t^*_{M-2} > \tilde{t}^*_{M-2}$. Then, $t^*_{M-3} > \tilde{t}^*_{M-3}$ (a smaller width is needed because the matching rates are higher and the rate is no larger, and the next value also increased). This shifting continues until $t^*_{k+1} > \tilde{t}^*_{k+1}$. Then, $t^*_k > \tilde{t}^*_k$. Suppose instead that the overall rate function increased when the matching function changed from $\tilde{g}$ to g . Then $g_1 < \tilde{g}_1$ and the target rate is larger, and so $t^*_1 > \tilde{t}^*_1$. Then, $t^*_2 > \tilde{t}^*_2$ (a larger width is needed and the previous value also increased). This shifting continues until $t^*_{k-1} > \tilde{t}^*_{k-1}$. Then, $t^*_k > \tilde{t}^*_k$.

Expanded PaperInterface specification

```
∀ {m k : ℕ},
  0 < k →
  ∀ (hkm : k < m + 1) (sampleRate shiftedRate levels shiftedLevels : Fin (m + 2) → ℝ),
  GJ190OptimalBinaryRatingSystems.BinaryEndpointLevelVector levels →
  GJ190OptimalBinaryRatingSystems.BinaryEndpointLevelVector shiftedLevels →
  GJ190OptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize levels sampleRate →
  GJ190OptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize shiftedLevels shiftedRate →
  (∀ (i : Fin (m + 2)), 0 < sampleRate i) →
  (∀ (i : Fin (m + 2)), 0 < shiftedRate i) →
  (∀ (i : Fin (m + 2)), k < i → shiftedRate i ≤ sampleRate i) →
  (∀ (i : Fin (m + 2)), i < k → sampleRate i ≤ shiftedRate i) →
  sampleRate (k, ...) = shiftedRate (k, ...) → shiftedLevels (k, ...) ≤ levels (k, ...)
```

Lean proof endpoint (built separately)

GJ190OptimalBinaryRatingSystems.PaperInterface.lemmaB1_matching_rate_shift

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 27

Source locator: cited publication, lines 1951-1979

Verbatim source input

Limit of β as $M \rightarrow \infty$

w denote the optimal β with M intervals for weight function w, with intervals $\{S_{wM}\} =$

Let β_M

i

w_M . Let q

w_M, s_{wM}), i.e. the quantile of

$\{s_i, s_{wM}\}$

)}

and

levels

t

(θ)

=

i/M

when

$\theta \in$

[s

w_M

i+1

i

i+1

interval item of type θ is in.

Then, we have the following convergence result for β_M .

Theorem B.1. Let g be uniform. Suppose w such that q_{wM} converges uniformly. Then, $\forall C \in$

Expanded PaperInterface specification

```
∀ (M : ℕ) (cellIndex : ℝ → Fin M) (levels : Fin M → ℝ) (quantile beta : ℝ → ℝ),
  (GJ190ptimalBinaryRatingSystems.ProofBridge.source_theoremB1_quantile_representation M cellIndex levels quantile
    beta =
      ∀ (θ : ℝ), quantile θ = ↑↑(cellIndex θ) / ↑M) ∧
  ∀ (θ : ℝ), beta θ = levels (cellIndex θ)
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_theoremB1_quantile_representation

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 28

Source locator: cited publication, lines 1980-2003

Verbatim source input

w
 w uniformly as $N \rightarrow \infty$.
 $N, \exists \beta$ w s.t. β^2
 $N+1 \rightarrow \beta$
The proof is technical and is below. We leverage the fact that, for g uniform, the levels of $\beta^2 M$ can be analytically written as a function of the levels of βM . We believe (numerically observe) that this theorem holds for the entire sequence as opposed to the each such subsequence, and for general matching functions g . However, our proof technique does not carry over, and the proof would leverage more global properties of the optimal βM .
Furthermore, the condition on w is light. For example, it holds for Kendall's τ , Spearman's ρ , and all other examples mentioned in this work.
This convergence result suggests that the choice of M when calculating a asymptotic and rate optimal β is not consequential. As M increases, the limiting value of W_k increases to 1 (i.e. the asymptotic value increases), but the optimal rate decreases to 0. As discussed above, with strictly increasing and continuous β , the asymptotic value is 1 but the large deviations rate does not exist, i.e. convergence is polynomial.
This result could potentially be strengthened as follows: first, show convergence on the entire sequence as opposed to these exponential subsequences, as conjectured; second, show desirable properties of the limiting function itself. It is conceivable but not necessarily true that the limiting

[form-feed in source extraction]
function is "better" than other strictly continuous increasing functions in some rate sense, even though the comparison through large deviations rate is degenerate.

Expanded PaperInterface specification

```
∀ (betaSeq quantileSeq : ℕ → ℝ → ℝ) (quantileLimit : ℝ → ℝ) (levels : (m : ℕ) → Fin (m + 2) → ℝ)
  (levelIndex : (m : ℕ) → ℝ → Fin (m + 2)),
  (∀ (m : ℕ) (θ : ℝ), betaSeq (m + 2) θ = levels m (levelIndex m θ)) →
  (∀ (m : ℕ),
    EconCSLib.Optimization.IsMaximizerOn GJ190OptimalBinaryRatingSystems.BinaryEndpointLevelVector
      (fun xs => GJ190OptimalBinaryRatingSystems.binaryEndpointAwareAdjacentRateObjective xs fun x => 1)
      (levels m)) →
  (∀ (m : ℕ), ∀ θ ∈ Set.Icc 0 1, ↑(levelIndex m θ) = min [↑(m + 2) * quantileSeq (m + 2) θ]₊ (m + 1)) →
  (∀ (m : ℕ), ∀ θ ∈ Set.Icc 0 1, quantileSeq (m + 2) θ ∈ Set.Icc 0 1) →
  TendstoUniformlyOn quantileSeq quantileLimit Filter.atTop (Set.Icc 0 1) →
  GJ190OptimalBinaryRatingSystems.theoremB1UniformOptimalSubsequencePrincipleTo betaSeq quantileSeq
  quantileLimit
```

Lean proof endpoint (built separately)

```
GJ190OptimalBinaryRatingSystems.PaperInterface.paper_theoremB1_uniform_subsequence_principle_to_of_quantile_floor_tendstoUniformly ]
↪ On_geometric_mesh
```

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 29

Source locator: cited publication, lines 2005-2027

Verbatim source input

Learning $\psi(\theta, y)$ through experiments

Now, we show how a platform would run an experiment to decide to learn $\psi(\theta, y)$. In particular, one potential issue is that the platform does not have any items with known quality that it can use as representative items in its optimization. In this case, we show that it can use ratings within the experiment itself to identify these representative items. The results essentially follow from the law of large numbers.

We assume that $|\Theta| = L$ representative items $i \in \{1 \dots L\}$ are in the experiment, and each are matched N times. The experiment proceeds as follows: every time an item is matched, show the buyer a random question from Y . For each word $y \in Y$, track the empirical $\psi(i, y)$, the proportion of times a positive response was given to question y . Alternatively, if Y is totally ordered (i.e. a positive rating for a given y also implies positive ratings would be given to all “easier” $y \preceq y$), and can be phrased as a multiple choice question, data collection can be faster: e.g., as we do in our experiments: Y consists of a set of totally ordered adjectives that can describe the item; the rater is asked to pick an adjective out of the set; this is interpreted as the item receiving a positive answer to the questions induced by the chosen answer and all worse adjectives, and a negative answer to all better adjectives.

First, suppose the platform approximately knows the quality θ_i of each item i , and θ_i are evenly distributed in $[0, 1]$. Suppose the items are ordered by index, i.e. $\theta_1 < \theta_2 < \dots < \theta_L$. Then let $\psi(\theta, y) = \psi(i, y)$ when $\theta \in [\theta_{i-1}, \theta_i]$. Call this procedure `KnownTypeExperiment`.

Lemma B.2. Suppose $\psi(\theta, y)$ is Lipschitz continuous in θ . With `KnownTypeExperiment`, $\psi(i, y) \rightarrow \psi(\theta_i, y) \forall y$ uniformly as $N \rightarrow \infty$. As $L \rightarrow \infty$, $\psi(\theta, y) \rightarrow \psi(\theta, y) \forall \theta$ uniformly.

Expanded PaperInterface specification

```
∀ (Representative : Type u_1) (Y : Type u_2) [inst : Fintype Representative] [inst_1 : Fintype Y]
  (value : GJ190ptimalBinaryRatingSystems.SourceKnownTypeExperiment Representative Y),
  ∃ quality empiricalResponse, value.quality = quality ∧ value.empiricalResponse = empiricalResponse
```

Lean proof endpoint (built separately)

`GJ190ptimalBinaryRatingSystems.PaperInterface.appendixB5_known_type_experiment`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 30

Source locator: cited publication, lines 2005-2027

Verbatim source input

Learning $\psi(\theta, y)$ through experiments

Now, we show how a platform would run an experiment to decide to learn $\psi(\theta, y)$. In particular, one potential issue is that the platform does not have any items with known quality that it can use as representative items in its optimization. In this case, we show that it can use ratings within the experiment itself to identify these representative items. The results essentially follow from the law of large numbers.

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First, suppose the platform approximately knows the quality θ_i of each item i , and θ_i are evenly distributed in $[0, 1]$. Suppose the items are ordered by index, i.e. $\theta_1 < \theta_2 < \dots < \theta_L$. Then let $\psi(\theta, y) = \psi(i, y)$ when $\theta \in [\theta_{i-1}, \theta_i]$. Call this procedure `KnownTypeExperiment`.

Lemma B.2. Suppose $\psi(\theta, y)$ is Lipschitz continuous in θ . With `KnownTypeExperiment`, $\psi(i, y) \rightarrow \psi(\theta_i, y) \forall y$ uniformly as $N \rightarrow \infty$. As $L \rightarrow \infty$, $\psi(\theta, y) \rightarrow \psi(\theta, y) \forall \theta$ uniformly.

Expanded PaperInterface specification

```
∀ (Ω : Type u_1) (Item : Type u_2) (Y : Type u_3)
  (value : GJ190ptimalBinaryRatingSystems.SourceRandomQuestionExperiment Ω Item Y),
  ∃ question positiveResponse, value.question = question ∧ value.positiveResponse = positiveResponse
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.appendixB5_random_question_experiment

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 31

Source locator: cited publication, lines 2005-2027

Verbatim source input

Learning $\psi(\theta, y)$ through experiments

Now, we show how a platform would run an experiment to decide to learn $\psi(\theta, y)$. In particular, one potential issue is that the platform does not have any items with known quality that it can use as representative items in its optimization. In this case, we show that it can use ratings within the experiment itself to identify these representative items. The results essentially follow from the law of large numbers.

We assume that $|\Theta| = L$ representative items $i \in \{1 \dots L\}$ are in the experiment, and each are matched N times. The experiment proceeds as follows: every time an item is matched, show the buyer a random question from Y . For each word $y \in Y$, track the empirical $\psi(i, y)$, the proportion of times a positive response was given to question y . Alternatively, if Y is totally ordered (i.e. a positive rating for a given y also implies positive ratings would be given to all “easier” $y' \leq y$), and can be phrased as a multiple choice question, data collection can be faster: e.g., as we do in our experiments: Y consists of a set of totally ordered adjectives that can describe the item; the rater is asked to pick an adjective out of the set; this is interpreted as the item receiving a positive answer to the questions induced by the chosen answer and all worse adjectives, and a negative answer to all better adjectives.

First, suppose the platform approximately knows the quality θ_i of each item i , and θ_i are evenly distributed in $[0, 1]$. Suppose the items are ordered by index, i.e. $\theta_1 < \theta_2 < \dots < \theta_L$. Then let $\psi(\theta, y) = \psi(i, y)$ when $\theta \in [\theta_{i-1}, \theta_i]$. Call this procedure `KnownTypeExperiment`.

Lemma B.2. Suppose $\psi(\theta, y)$ is Lipschitz continuous in θ . With `KnownTypeExperiment`, $\psi(i, y) \rightarrow \psi(\theta_i, y) \forall y$ uniformly as $N \rightarrow \infty$. As $L \rightarrow \infty$, $\psi(\theta, y) \rightarrow \psi(\theta, y) \forall \theta$ uniformly.

Expanded PaperInterface specification

```
∀ {Ω : Type u_1} {Item : Type u_2} {Y : Type u_3} [inst : DecidableEq Y]
  (E : GJ190ptimalBinaryRatingSystems.SourceRandomQuestionExperiment Ω Item Y) (i : Item) (y : Y) (N : ℕ) (ω : Ω),
  GJ190ptimalBinaryRatingSystems.sourceExperimentEmpiricalQuestionResponse E i y N ω =
    GJ190ptimalBinaryRatingSystems.sourceExperimentPositiveQuestionCount E i y N ω /
    GJ190ptimalBinaryRatingSystems.sourceExperimentQuestionCount E i y N ω
```

Lean proof endpoint (built separately)

`GJ190ptimalBinaryRatingSystems.PaperInterface.appendixB5_empirical_question_response`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 32

Source locator: cited publication, lines 2027-2031

Verbatim source input

$\psi(\theta_i, y) \forall y$ uniformly as $N \rightarrow \infty$. As $L \rightarrow \infty$, $\psi(\theta, y) \rightarrow \psi(\theta, y) \forall \theta$ uniformly.
Proof. The proof follows directly from the Strong Law of Large Numbers. As $N \rightarrow \infty$, $\forall i$, $\psi(i, x) \rightarrow \psi(\theta_i, x)$ uniformly. Now, let $L \rightarrow \infty$. $\forall [U+000F \text{ control character}]$, $\exists L_0$ s.t. $\forall L > L_0$, $\forall \theta$, $\exists i$ s.t. $|\theta - \theta_i| < [U+000F \text{ control character}]$. $\psi(\theta, x)$ is Lipschitz in θ by assumption, and so $\psi(\theta, x) \rightarrow \psi(\theta, x)$ uniformly.
We now relax the assumption that the platform has an existing set of items with known qualities.

Expanded PaperInterface specification

```

∀ {Ω : Type u_1} {Y : Type u_2} [inst : MeasurableSpace Ω] [Fintype Y] [inst_2 : DecidableEq Y]
(P : MeasureTheory.Measure Ω)
(E : (L : ℕ) → GJ190ptimalBinaryRatingSystems.SourceRandomQuestionExperiment Ω (Fin (L + 1)) Y)
(quality : (L : ℕ) → Fin (L + 1) → ℝ) (representative : (L : ℕ) → ℝ → Fin (L + 1)) (psi : ℝ → Y → ℝ) (H : Y → ℝ),
(∀ (y : Y), 0 < H y) →
  (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
    MeasureTheory.Integrable (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator (E L) i y 0) P) →
    (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
      Pairwise
        (Function.onFun (fun x1 x2 => ProbabilityTheory.IndepFun x1 x2 P)
          (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator (E L) i y))) →
    (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y) (n : ℕ),
      ProbabilityTheory.IdentDistrib (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator (E L) i y n)
        (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator (E L) i y 0) P P) →
    (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
      ∫ (ω : Ω), GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator (E L) i y 0 ω ∂P = H y) →
    (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
      MeasureTheory.Integrable (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator (E L) i y 0)
        P) →
    (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
      Pairwise
        (Function.onFun (fun x1 x2 => ProbabilityTheory.IndepFun x1 x2 P)
          (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator (E L) i y))) →
    (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y) (n : ℕ),
      ProbabilityTheory.IdentDistrib
        (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator (E L) i y n)
        (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator (E L) i y 0) P P) →
    (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
      ∫ (ω : Ω), GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator (E L) i y 0 ω ∂P =
        H y * psi (quality L i) y) →
    (∀ δ > 0,
      ∀r (L : ℕ) in Filter.atTop, ∀ θ ∈ Set.Icc 0 1, dist (quality L (representative L θ)) θ < δ) →
      ∀ {K : ℝ},
        0 ≤ K →
        (∀ (θ θ' : ℝ) (y : Y), dist (psi θ y) (psi θ' y) ≤ K * dist θ θ') →
        ∀n (ω : Ω) ∂P,
          ∀ ε > 0,
            ∀r (L : ℕ) (N : ℕ) in Filter.atTop,
              ∀ θ ∈ Set.Icc 0 1,
                ∀ (y : Y),
                  dist
                    (GJ190ptimalBinaryRatingSystems.sourceExperimentEmpiricalQuestionResponse
                      (E L) (representative L θ) y N ω)
                    (psi θ y) <
                      ε
    )
  )

```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.paper_lemmaB2_knownTypeExperiment_uniform_convergence_of_lipschitz_tracking

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 33

Source locator: cited publication, lines 2005-2031

Verbatim source input

Learning $\psi(\theta, y)$ through experiments

Now, we show how a platform would run an experiment to decide to learn $\psi(\theta, y)$. In particular, one potential issue is that the platform does not have any items with known quality that it can use as representative items in its optimization. In this case, we show that it can use ratings within the experiment itself to identify these representative items. The results essentially follow from the law of large numbers.

We assume that $|\Theta| = L$ representative items $i \in \{1 \dots L\}$ are in the experiment, and each are matched N times. The experiment proceeds as follows: every time an item is matched, show the buyer a random question from Y . For each word $y \in Y$, track the empirical $\psi(i, y)$, the proportion of times a positive response was given to question y . Alternatively, if Y is totally ordered (i.e. a positive rating for a given y also implies positive ratings would be given to all “easier” $y \theta$), and can be phrased as a multiple choice question, data collection can be faster: e.g., as we do in our experiments: Y consists of a set of totally ordered adjectives that can describe the item; the rater is asked to pick an adjective out of the set; this is interpreted as the item receiving a positive answer to the questions induced by the chosen answer and all worse adjectives, and a negative answer to all better adjectives.

First, suppose the platform approximately knows the quality θ_i of each item i , and θ_i are evenly distributed in $[0, 1]$. Suppose the items are ordered by index, i.e. $\theta_1 < \theta_2 < \dots < \theta_L$. Then let $\psi(\theta, y) = \psi(i, y)$ when $\theta \in [\theta_{i-1}, \theta_i]$. Call this procedure `KnownTypeExperiment`.

Lemma B.2. Suppose $\psi(\theta, y)$ is Lipschitz continuous in θ . With `KnownTypeExperiment`, $\psi(i, y) \rightarrow \psi(\theta_i, y) \forall y$ uniformly as $N \rightarrow \infty$. As $L \rightarrow \infty$, $\psi(\theta, y) \rightarrow \psi(\theta, y) \forall \theta$ uniformly.

Proof. The proof follows directly from the Strong Law of Large Numbers. As $N \rightarrow \infty$, $\forall i$, $\psi(i, x) \rightarrow \psi(\theta_i, x)$ uniformly. Now, let $L \rightarrow \infty$. $\forall [U+\epsilon]F$ control character, $\exists L_0$ s.t. $\forall L > L_0$, $\forall \theta$, $\exists i$ s.t. $|\theta - \theta_i| < [U+\epsilon]F$ control character. $\psi(\theta, x)$ is Lipschitz in θ by assumption, and so $\psi(\theta, x) \rightarrow \psi(\theta, x)$ uniformly.

We now relax the assumption that the platform has an existing set of items with known qualities.

Expanded PaperInterface specification

```
∀ {Ω : Type u_1} {Representative : Type u_2} {Y : Type u_3} [inst : MeasurableSpace Ω] [Fintype Representative]
[Fintype Y] [inst_3 : DecidableEq Y] (P : MeasureTheory.Measure Ω)
(E : GJ190ptimalBinaryRatingSystems.SourceRandomQuestionExperiment Ω Representative Y) (quality : Representative → ℝ)
(ψ : ℝ → Y → ℝ) (H : Y → ℝ),
(∀ (y : Y), 0 < H y) →
(∀ (i : Representative) (y : Y),
  MeasureTheory.Integrable (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator E i y 0) P) →
(∀ (i : Representative) (y : Y),
  Pairwise
    (Function.onFun (fun x1 x2 => ProbabilityTheory.IndepFun x1 x2 P)
      (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator E i y))) →
(∀ (i : Representative) (y : Y) (n : ℕ),
  ProbabilityTheory.IdentDistrib (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator E i y n)
    (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator E i y 0) P P) →
(∀ (i : Representative) (y : Y),
  ∫ (ω : Ω), GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator E i y 0 ω ∂P = H y) →
(∀ (i : Representative) (y : Y),
  MeasureTheory.Integrable (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator E i y 0) P) →
(∀ (i : Representative) (y : Y),
  Pairwise
    (Function.onFun (fun x1 x2 => ProbabilityTheory.IndepFun x1 x2 P)
      (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator E i y))) →
(∀ (i : Representative) (y : Y) (n : ℕ),
  ProbabilityTheory.IdentDistrib
    (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator E i y n)
    (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator E i y 0) P P) →
(∀ (i : Representative) (y : Y),
  ∫ (ω : Ω), GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator E i y 0 ω ∂P =
    H y * ψ (quality i) y) →
∀ (ω : Ω) ∂P,
  ∀ (i : Representative) (y : Y),
  Filter.Tendsto
    (fun N => GJ190ptimalBinaryRatingSystems.sourceExperimentEmpiricalQuestionResponse E i y N ω)
    Filter.atTop (nhds (ψ (quality i) y))
```

Lean proof endpoint (built separately)

`GJ190ptimalBinaryRatingSystems.PaperInterface.lemmaB2_knownTypeExperiment_random_question_slln`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 34

Source locator: cited publication, lines 2032-2044

Verbatim source input

Suppose instead the platform has many items L of unknown quality who are expected to match N times each over the experiment time period. For each item, the platform would again ask questions from Y , drawn according to any distribution (with positive mass on each question). Then generate $\psi(\theta, y)$ as follows: first, rank the items according to their ratings during the experiment itself. Then, for each y , $\psi(\theta, y)$ is the empirical performance of the θ th percentile item in the ranking, i.e.

```
[U+0002 control character] i-1
[U+0003 control character]
i
 $\psi(\theta, y) = \psi(\theta_i, y)$  for  $\theta \in L, L$ . Call this procedure UnknownTypeExperiment.
```

Lemma B.3. Suppose $\psi(\theta, y)$ is Lipschitz continuous in θ . With `UnknownTypeExperiment`, $\psi(\theta, y) \rightarrow \psi(\theta, y) \forall y$, θ uniformly as $L, N \rightarrow \infty$.

Expanded PaperInterface specification

```
∀ (Item : Type u_1) (Y : Type u_2) [inst : Fintype Item] [inst_1 : Fintype Y]
  (value : GJ190ptimalBinaryRatingSystems.SourceUnknownTypeExperiment Item Y),
  ∃ trueQuality empiricalResponse empiricalAverageScore rankedItem,
    value.trueQuality = trueQuality ∧
    value.empiricalResponse = empiricalResponse ∧
    value.empiricalAverageScore = empiricalAverageScore ∧ value.rankedItem = rankedItem
```

Lean proof endpoint (built separately)

`GJ190ptimalBinaryRatingSystems.PaperInterface.appendixB5_unknown_type_experiment`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 35

Source locator: cited publication, lines 2044-2060

Verbatim source input

$\psi(\theta, y) \forall y, \theta$ uniformly as $L, N \rightarrow \infty$.
Proof. Fix L . Denote each item in the experiment as $i \in \{1 \dots L\}$ (with true quality $\theta_i \neq \theta_j$), and each item has N samples. Without loss of generality, assume the items are indexed according to their rank on the average of their scores on the samples, defined as the percentage of positive ratings received. $i = 1$ is then the worst item, and $i = L$ is the best item according to scores in the experiment.
For $\psi(\theta, x)$ increasing in θ , as $N \rightarrow \infty$, $P(r(\theta_i > \theta_j | i < j) \rightarrow \theta)$ almost surely by SLLN, and for a fixed L , $\{\theta_i\}$ this convergence [U+0002 control character] is uniform.
Furthermore, by SLLN, $\psi(i, x) \rightarrow \psi(\theta_i, x)$ as $N \rightarrow \infty$.
[U+0003 control character]
i-1 i
Recall $\psi(\theta, x) = \psi(i, x)$ for $\theta \in L, L$.
10

[form-feed in source extraction]
Now, let $L \rightarrow \infty$. $\forall [U+000F control character], \exists L_0$ s.t. $\forall L > L_0, \forall \theta, \exists i$ s.t. $|\theta - \theta_i| < [U+000F control character]$. $\psi(\theta, x)$ is $\hookrightarrow$ Lipschitz in θ by assumption, and so $\psi(\theta, x) \rightarrow \psi(\theta, x)$ uniformly.

Expanded PaperInterface specification

```

V {Ω : Type u_1} {Y : Type u_2} [inst : MeasurableSpace Ω] [inst_1 : Fintype Y] [inst_2 : DecidableEq Y]
(P : MeasureTheory.Measure Ω)
(E : (L : ℕ) → GJ190ptimalBinaryRatingSystems.SourceRandomQuestionExperiment Ω (Fin (L + 1)) Y)
(quality : (L : ℕ) → Fin (L + 1) → ℝ) (rankRepresentative : (L : ℕ) → ℝ → Fin (L + 1)) (psi : ℝ → Y → ℝ) (H : Y → ℝ),
(∀ (y : Y), 0 < H y) →
  (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
    MeasureTheory.Integrable (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator (E L) i y 0) P) →
    (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
      Pairwise
        (Function.onFun (fun x1 x2 => ProbabilityTheory.IndepFun x1 x2 P)
          (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator (E L) i y))) →
      (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y) (n : ℕ),
        ProbabilityTheory.IdentDistrib (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator (E L) i y n)
          (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator (E L) i y 0) P P) →
      (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
        ∫ (ω : Ω), GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator (E L) i y 0 ω ∂P = H y) →
      (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
        MeasureTheory.Integrable (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator (E L) i y 0)
          P) →
      (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
        Pairwise
          (Function.onFun (fun x1 x2 => ProbabilityTheory.IndepFun x1 x2 P)
            (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator (E L) i y))) →
      (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y) (n : ℕ),
        ProbabilityTheory.IdentDistrib
          (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator (E L) i y n)
          (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator (E L) i y 0) P P) →
      (∀ (L : ℕ) (i : Fin (L + 1)) (y : Y),
        ∫ (ω : Ω), GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator (E L) i y 0 ω ∂P =
          H y * psi (quality L i) y) →
      (∀ (L : ℕ) (i : Fin (L + 1)),
        MeasureTheory.Integrable
          (GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator (E L) i 0) P) →
      (∀ (L : ℕ) (i : Fin (L + 1)),
        Pairwise
          (Function.onFun (fun x1 x2 => ProbabilityTheory.IndepFun x1 x2 P)
            (GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator (E L) i))) →
      (∀ (L : ℕ) (i : Fin (L + 1)) (n : ℕ),
        ProbabilityTheory.IdentDistrib
          (GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator (E L) i n)
          (GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator (E L) i 0) P P) →
      (∀ (L : ℕ) (i : Fin (L + 1)),
        ∫ (ω : Ω),
          GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator (E L) i 0 ω ∂P =
          GJ190ptimalBinaryRatingSystems.sourceExpectedAggregateScore psi H (quality L i)) →
      (∀ (L : ℕ),
        StrictMono fun i =>
          GJ190ptimalBinaryRatingSystems.sourceExpectedAggregateScore psi H (quality L i)) →
      (∀ δ > 0,
        ∀f (L : ℕ) in Filter.atTop,
          ∀ θ ∈ Set.Icc 0 1, dist (quality L (rankRepresentative L θ)) θ < δ) →
        ∀ {K : ℝ},
          0 ≤ K →
          (∀ (θ θ' : ℝ) (y : Y), dist (psi θ y) (psi θ' y) ≤ K * dist θ θ') →
          ∀m (ω : Ω) ∂P,
            ∀ ε > 0,
              ∀f (L : ℕ) (N : ℕ) in Filter.atTop,
                (StrictMono fun i =>
                  GJ190ptimalBinaryRatingSystems.sourceExperimentEmpiricalMean
                    (GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator (E L)
                      i)
                    N ω) ∧
                  ∀ θ ∈ Set.Icc 0 1,
                    ∀ (y : Y),
                      dist

```

$$\begin{aligned} & (GJ190ptimalBinaryRatingSystems.sourceExperimentEmpiricalQuestionResponse \\ & \quad (E \ L) \ (rankRepresentative \ L \ \theta) \ y \ N \ \omega) \\ & \quad (\psi \ \theta \ y) < \\ & \epsilon \end{aligned}$$

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.paper_lemmaB3_unknownTypeExperiment_uniform_convergence_of_ranking_tracking

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 36

Source locator: cited publication, lines 2032-2060

Verbatim source input

Suppose instead the platform has many items L of unknown quality who are expected to match N times each over the experiment time period. For each item, the platform would again ask questions from Y , drawn according to any distribution (with positive mass on each question). Then generate $\psi(\theta, y)$ as follows: first, rank the items according to their ratings during the experiment itself. Then, for each y , $\psi(\theta, y)$ is

the empirical performance of the θ th percentile item in the ranking, i.e.

[U+0002 control character] $i-1$

[U+0003 control character]

i

$\psi(\theta, y) = \psi(\theta_i, y)$ for $\theta \in L$, L . Call this procedure `UnknownTypeExperiment`.

Lemma B.3. Suppose $\psi(\theta, y)$ is Lipschitz continuous in θ . With `UnknownTypeExperiment`, $\psi(\theta, y) \rightarrow$

$\psi(\theta, y) \forall y, \theta$ uniformly as $L, N \rightarrow \infty$.

Proof. Fix L . Denote each item in the experiment as $i \in \{1 \dots L\}$ (with true quality $\theta_i \neq \theta_j$), and each item has N samples. Without loss of generality, assume the items are indexed according to their rank on the average of their scores on the samples, defined as the percentage of positive ratings received. $i = 1$ is then the worst item, and $i = L$ is the best item according to scores in the experiment.

For $\psi(\theta, x)$ increasing in θ , as $N \rightarrow \infty$, $P(r(\theta_i > \theta_j) | i < j) \rightarrow \theta$ almost surely by SLLN, and for a fixed L , $\{\theta_i\}$ this convergence [U+0002 control character] is uniform.

Furthermore, by SLLN, $\psi(i, x) \rightarrow \psi(\theta_i, x)$ as $N \rightarrow \infty$.

[U+0003 control character]

$i-1$

Recall $\psi(\theta, x) = \psi(i, x)$ for $\theta \in L$, L .

$\square$

[form-feed in source extraction]

Now, let $L \rightarrow \infty$. $\forall$ [U+000F control character], $\exists L_0$ s.t. $\forall L > L_0$, $\forall \theta, \exists i$ s.t. $|\theta - \theta_i| < [U+000F control character]$. $\psi(\theta, x)$ is

$\hookrightarrow$ Lipschitz in θ by

assumption, and so $\psi(\theta, x) \rightarrow \psi(\theta, x)$ uniformly.

Expanded PaperInterface specification

```

∀ {Ω : Type u_1} {Y : Type u_2} {L : ℕ} [inst : MeasurableSpace Ω] [inst_1 : Fintype Y] [inst_2 : DecidableEq Y]
(P : MeasureTheory.Measure Ω) (E : GJ190ptimalBinaryRatingSystems.SourceRandomQuestionExperiment Ω (Fin L) Y)
(quality : Fin L → ℝ) (ψ : ℝ → Y → ℝ) (H : Y → ℝ),
(∀ (y : Y), 0 < H y) →
(∀ (i : Fin L) (y : Y),
  MeasureTheory.Integrable (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator E i y 0) P) →
(∀ (i : Fin L) (y : Y),
  Pairwise
    (Function.onFun (fun x1 x2 => ProbabilityTheory.IndepFun x1 x2 P)
      (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator E i y))) →
(∀ (i : Fin L) (y : Y) (n : ℕ),
  ProbabilityTheory.IdentDistrib (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator E i y n)
    (GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator E i y 0) P P) →
(∀ (i : Fin L) (y : Y),
  ∫ (ω : Ω), GJ190ptimalBinaryRatingSystems.sourceQuestionAskedIndicator E i y 0 ω ∂P = H y) →
(∀ (i : Fin L) (y : Y),
  MeasureTheory.Integrable (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator E i y 0) P) →
(∀ (i : Fin L) (y : Y),
  Pairwise
    (Function.onFun (fun x1 x2 => ProbabilityTheory.IndepFun x1 x2 P)
      (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator E i y))) →
(∀ (i : Fin L) (y : Y) (n : ℕ),
  ProbabilityTheory.IdentDistrib
    (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator E i y n)
    (GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator E i y 0) P P) →
(∀ (i : Fin L) (y : Y),
  ∫ (ω : Ω), GJ190ptimalBinaryRatingSystems.sourceQuestionPositiveIndicator E i y 0 ω ∂P =
    H y * ψ (quality i) y) →
(∀ (i : Fin L),
  MeasureTheory.Integrable (GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator E i 0)
    P) →
(∀ (i : Fin L),
  Pairwise
    (Function.onFun (fun x1 x2 => ProbabilityTheory.IndepFun x1 x2 P)
      (GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator E i))) →
(∀ (i : Fin L) (n : ℕ),
  ProbabilityTheory.IdentDistrib
    (GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator E i n)
    (GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator E i 0) P P) →
(∀ (i : Fin L),
  ∫ (ω : Ω), GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator E i 0 ω ∂P =
    GJ190ptimalBinaryRatingSystems.sourceExpectedAggregateScore ψ H (quality i)) →
(StrictMono fun i =>
  GJ190ptimalBinaryRatingSystems.sourceExpectedAggregateScore ψ H (quality i)) →
  v^m (ω : Ω) ∂P,
  TendstoUniformlyOn
    (fun N p =>
      GJ190ptimalBinaryRatingSystems.sourceExperimentEmpiricalQuestionResponse E p.1 p.2
        N ω)
    (fun p => ψ (quality p.1) p.2) Filter.atTop Set.univ ∧
  ∀ f (N : ℕ) in Filter.atTop,
  StrictMono fun i =>
    GJ190ptimalBinaryRatingSystems.sourceExperimentEmpiricalMean
      (GJ190ptimalBinaryRatingSystems.sourceAggregatePositiveIndicator E i) N ω

```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.lemmaB3_unknownTypeExperiment_random_question_response_and_rank

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 37

Source locator: cited publication, lines 2083-2131

Verbatim source input

Rate functions for $P_k(\theta_1, \theta_2)$

```
Lemma C.1.
1
lim - log [μ((xk (θ1) - xk (θ2)) ≤ 0|θ1, θ2)] = inf {g(θ1)I(a|θ1) + g(θ2)I(a|θ2)}
a∈R
k→∞ k
where I(a|θ) = supz {za - Λ(z|θ)}, and Λ(z|θ) is the log moment generating function of a single
sample from x(θ1) and g(θ) is the sampling rate.
Proof. limk→∞ - k1 log [μ((xk (θ1) - xk (θ2)) ≤ 0|θ1, θ2)]
[U+0015 control character]
[U+0014 control character]Z
1
μ((xk (θ1) = a|θ1)μ(xk (θ2) ≥ a|θ2) da
= lim - log
k→∞ k
[U+0014 control character]Za∈R
[U+0015 control character]
1
-k g(θ1)I(a|θ1) - k g(θ2)I(a|θ2)
= lim - log
e
e
da
k→∞ k
a∈R
= inf {g(θ1)I(a|θ1) + g(θ2)I(a|θ2)}

(1)
(2)
Laplace principle

a∈R

(3)
```

Where (2) is a basic result from large deviations, where $kg(\theta_i)$ is the number of samples item of quality θ_i has received.

Note that this lemma also appears in Glynn and Juneja (2004), which uses the Gartner-Ellis Theorem in the proof. Our proof is conceptually similar but instead uses Laplace's principle.

```
1-b
Recall that  $KL(a||b) = a \log ab + (1 - a) \log 1-a$ 
is the Kullback-Leibler (KL) divergence between
Bernoulli random variables with success probabilities  $a$  and  $b$  respectively. It is well known that
for a Bernoulli random variable with success probability  $t$ ,
 $I(a|t) = KL(a||t)$ 
11
```

Expanded PaperInterface specification

```
∀ {Seller : Type u_1} {Pair : Type u_2} (successProb : Seller → ℝ) (hprob0 : ∀ (θ : Seller), 0 ≤ successProb θ)
(hprob1 : ∀ (θ : Seller), successProb θ ≤ 1),
(∀ (θ : Seller), 0 < successProb θ) →
(∀ (θ : Seller), successProb θ < 1) →
  ∀ (sampleRate : Seller → ℝ) (pairHi pairLo : Pair → Seller),
  (∀ (p : Pair), 0 < sampleRate (pairHi p)) →
  (∀ (p : Pair), 0 < sampleRate (pairLo p)) →
    ∀ (a z : Pair → ℝ),
    (∀ (p : Pair), z p ≤ 0) →
      (∀ (p : Pair),
        HasDerivAt
          (fun t =>
            (GJ190ptimalBinaryRatingSystems.binaryRatingModel successProb hprob0 hprob1).logMGF (pairHi p)
              t)
          (a p) (z p * (sampleRate (pairHi p))⁻¹)) →
        (∀ (p : Pair),
          HasDerivAt
            (fun t =>
              (GJ190ptimalBinaryRatingSystems.binaryRatingModel successProb hprob0 hprob1).logMGF (pairLo p)
                t)
            (a p) (-(z p * (sampleRate (pairLo p))⁻¹))) →
          Nonempty
            (EconCSLib.Probability.PairwiseThresholdRateTopLdpCertificate
              (GJ190ptimalBinaryRatingSystems.binaryRatingModel successProb hprob0 hprob1) sampleRate pairHi
                pairLo)
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_lemmaC1_pairwise_error_rate_from_derivatives

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 38

Source locator: cited publication, lines 2133-2161

Verbatim source input

```
Lemma C.2. Let  $\theta_1 > \theta_2$  and  $I(a|\theta) = KL(a||\beta(\theta))$ . Further, Let  $P_k(\theta_1, \theta_2) = 1 - P_k(\theta_1, \theta_2)$ .  
Then,  
1  
-  $\lim_{k \rightarrow \infty} \log P_k(\theta_1, \theta_2) = \inf_{a \in \mathbb{R}} \{g(\theta_1)I(a|\theta_1) + g(\theta_2)I(a|\theta_2)\}$ ,  
(4)  
a  $\in \mathbb{R}$   
k  $\rightarrow \infty$   
Proof. Follows directly from Lemma C.1.  
-  $\lim_{k \rightarrow \infty}$   
1  
k  $\rightarrow \infty$   
log  $P_k(\theta_1, \theta_2 | \beta)$   
1  
=  $\lim_{k \rightarrow \infty} -\log [1 + \mu_k(x_k(\theta_1) - x_k(\theta_2) < 0 | \theta_1, \theta_2) - \mu_k(x_k(\theta_1) - x_k(\theta_2) > 0 | \theta_1, \theta_2)]$   
k  $\rightarrow \infty$   
1  
=  $\lim_{k \rightarrow \infty} -\log [2\mu_k(x_k(\theta_1) - x_k(\theta_2) < 0 | \theta_1, \theta_2) + \mu_k(x_k(\theta_1) - x_k(\theta_2) = 0 | \theta_1, \theta_2)]$   
k  $\rightarrow \infty$   
1  
=  $\lim_{k \rightarrow \infty} -\log [\mu_k(x_k(\theta_1) - x_k(\theta_2) \leq 0 | \theta_1, \theta_2)]$   
k  $\rightarrow \infty$   
=  $\inf_{a \in \mathbb{R}} \{g(\theta_1)I(a|\theta_1) + g(\theta_2)I(a|\theta_2)\}$   
Lemma C.1  
a  $\in \mathbb{R}$ 
```

Expanded PaperInterface specification

```
∀ {Seller : Type u_1} (successProb : Seller → ℝ) (hprob0 : ∀ (θ : Seller), 0 ≤ successProb θ)  
(hprob1 : ∀ (θ : Seller), successProb θ ≤ 1) (sampleRate : Seller → ℝ) (hi lo : Seller),  
  θ < sampleRate hi →  
    θ < sampleRate lo →  
      θ < successProb hi →  
        successProb hi < 1 →  
          θ < successProb lo →  
            successProb lo < 1 →  
              successProb lo ≤ successProb hi →  
                EconCSLib.Probability.ExponentialRateCertificate  
                  (EconCSLib.Probability.twoSampleFloorPkComplementErrorProb  
                    (GJ190ptimalBinaryRatingSystems.binaryRatingModel successProb hprob0 hprob1) sampleRate hi lo)  
                    (EconCSLib.Probability.weightedBernoulliClosedThresholdRate (sampleRate hi) (sampleRate lo)  
                      (successProb hi) (successProb lo)))
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_lemmaC2_binary_complement_rate

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 39

Source locator: cited publication, lines 2163-2223

Verbatim source input

Laplace's principle with sequence of rate functions

In order to derive a rate function for $W_k = (\lim W_k) - W_k$, we need to be able to relate its rate to that of $P_k(\theta_1, \theta_2)$. The following theorem, related to Laplace's principle for large deviations allows us to do so.

Theorem C.1. Suppose that X is compact with finite Lebesgue measure $\mu(X) < \infty$. Suppose that $\phi(x)$ has an essential infimum ϕ on X , that $\phi_n(x)$ has an essential infimum ϕ_n , that both ϕ and all ϕ_n are nonnegative, and that $\phi_n \rightarrow \phi$ uniformly:
 $\limsup |\phi_n(x) - \phi(x)| = 0$.

$n \rightarrow \infty \quad x \in X$

Then:

$\lim_{n \rightarrow \infty} \log \int_X e^{-n\phi_n(x)} dx = -\phi$.

(5)

X

Proof. First, we note that for all x and n , $e^{-n\phi_n(x)} \leq e^{-n\phi}$. Therefore, letting $(*)$ denote the LHS of (5), we have:

X

$\lim_{n \rightarrow \infty} \log \int_X e^{-n\phi_n(x)} dx = -\phi$,

where the last limit follows from the fact that ϕ_n converges uniformly to ϕ , so that $\phi_n \rightarrow \phi$. Next, for $\epsilon > 0$ let $A_n(\epsilon) = \{x : \phi_n(x) \leq \phi + \epsilon\}$ and $A(\epsilon) = \{x : \phi(x) \leq \phi + \epsilon\}$. It follows (again by uniform convergence) that for all sufficiently large n , $A(\epsilon/2) \subseteq A_n(\epsilon)$, so that $\mu(A_n(\epsilon/2)) \leq \mu(A_n(\epsilon))$ for all sufficiently large n . Further, $\mu(A(\epsilon/2)) > 0$, since ϕ is the essential infimum.

Since:

X

$\int_X e^{-n\phi_n(x)} dx \geq \mu(A_n(\epsilon/2)) e^{-n(\phi + \epsilon/2)}$,
 X

$\lim_{n \rightarrow \infty} \log \int_X e^{-n\phi_n(x)} dx \geq -\phi - \epsilon$.

To complete the proof, observe that since $\mu(A_n(\epsilon))$ is bounded below by a positive constant for all sufficiently large n , the last limit is zero. Therefore:

$(*) \geq -\phi - \epsilon$. Since ϵ was arbitrary, this completes the proof.

Remark C.1. Let $X = [0, 1] \times [0, 1]$, $\phi(\theta_1, \theta_2) = -n \log P_n(\theta_1, \theta_2)$. Then, all the conditions for

X

$\int_X e^{-n\phi_n(x)} dx \geq \mu(A_n(\epsilon/2)) e^{-n(\phi + \epsilon/2)}$,
 X

12

[form-feed in source extraction]

it follows that:

$\lim_{n \rightarrow \infty} \log \mu(A_n(\epsilon/2)) \geq -\phi - \epsilon$.

To complete the proof, observe that since $\mu(A_n(\epsilon))$ is bounded below by a positive constant for all sufficiently large n , the last limit is zero. Therefore:

$(*) \geq -\phi - \epsilon$.

Since ϵ was arbitrary, this completes the proof.

Remark C.1. Let $X = [0, 1] \times [0, 1]$, $\phi(\theta_1, \theta_2) = -n \log P_n(\theta_1, \theta_2)$. Then, all the conditions for

X

Expanded PaperInterface specification

```
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] (μ : MeasureTheory.Measure Ω) [MeasureTheory.IsFiniteMeasure μ]
  (phiSeq : ℕ → Ω → ℝ) (phi : Ω → ℝ) {rate : ℝ},
  (∀ (k : ℕ), MeasureTheory.Integrable (fun x => Real.exp (-↑k * phiSeq k x)) μ) →
  EconCSLib.Probability.HasAEEssentialInfimum μ phi rate →
  (∀ ε > 0, ∀ (k : ℕ) in Filter.atTop, ∀ (x : Ω), |phiSeq k x - phi x| ≤ ε) →
  EconCSLib.Probability.HasExponentialRate (fun k => ∫ (x : Ω), Real.exp (-↑k * phiSeq k x) ∂μ) rate
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_theoremC1_laplace_principle

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 40

Source locator: cited publication, lines 2224-2225

Verbatim source input

Theorem C.1 are met.

Expanded PaperInterface specification

```
∀ (weight : ℝ × ℝ → ℝ) (phiSeq : ℕ → ℝ × ℝ → ℝ) (phi : ℝ × ℝ → ℝ) {rate W : ℝ},
  (∀ (k : ℕ),
    MeasureTheory.Integrable (fun x => weight x * Real.exp (-↑k * phiSeq k x))
      ((MeasureTheory.volume.restrict (Set.Icc 0 1)).prod (MeasureTheory.volume.restrict (Set.Icc 0 1)))) →
  0 < W →
  (∀m (x : ℝ × ℝ) ∂(MeasureTheory.volume.restrict (Set.Icc 0 1)).prod (MeasureTheory.volume.restrict (Set.Icc 0 1)),
    0 ≤ weight x) →
  (∀m (x :
    ℝ × ℝ) ∂(MeasureTheory.volume.restrict (Set.Icc 0 1)).prod (MeasureTheory.volume.restrict (Set.Icc 0 1)),
    weight x ≤ W) →
  EconCSLib.Probability.HasAEEssentialInfimum
    ((MeasureTheory.volume.restrict (Set.Icc 0 1)).prod (MeasureTheory.volume.restrict (Set.Icc 0 1))) phi
    rate →
  EconCSLib.Probability.HasPositiveWeightNearAEEssentialInfimum
    ((MeasureTheory.volume.restrict (Set.Icc 0 1)).prod (MeasureTheory.volume.restrict (Set.Icc 0 1)))
    weight phi rate →
  (∀ ε > 0, ∀f (k : ℕ) in Filter.atTop, ∀ (x : ℝ × ℝ), |phiSeq k x - phi x| ≤ ε) →
  EconCSLib.Probability.HasExponentialRate
    (fun k =>
      ∫ (x : ℝ × ℝ),
        weight x *
        Real.exp
          (-↑k *
            phiSeq k
              x) ∂(MeasureTheory.volume.restrict (Set.Icc 0 1)).prod
            (MeasureTheory.volume.restrict (Set.Icc 0 1)))
    rate
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_remarkC1_full_square_application_under_explicit_conditions

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 41

Source locator: cited publication, lines 2227-2364

Verbatim source input

Rate function for W_k

Our next lemma shows that we can obtain a nontrivial large deviations rate for W_k when β is a step-wise increasing

R function.

Recall $W_k = \theta_1 > \theta_2 \dots \theta_k \sum_{i=1}^k P_k(\theta_1, \theta_2 | \beta) d(\theta_1, \theta_2)$.

Let $P_k(\theta_1, \theta_2) = 1 - P_k(\theta_1, \theta_2)$.

R

Further, let $W_k = (\lim_{k \rightarrow \infty} W_k) - W_k = \theta_1 > \theta_2 \dots \theta_k \sum_{i=1}^k P_k(\theta_1, \theta_2 | \beta) d(\theta_1, \theta_2)$. (recall we assumed w integrates to 1 without loss of generality).

Lemma C.3. Suppose β is piecewise constant with M levels $\{t_i\}$. Let $g_i, \inf_{\theta \in S_i} g(\theta) = g(s_i)$

Then,

$$\lim_{k \rightarrow \infty} \log W_k = \min_{\theta_1 \leq i \leq M} \inf_{a \in R} \{g_{i+1} I(a|t_{i+1}) + g_i I(a|t_i)\}, \quad (6)$$

$\theta_1 \leq i \leq M, a \in R$

$k \rightarrow \infty$

where $I(a|t) = KL(a||t)$ as defined in Lemma C.2.

Proof. When β is step-wise increasing with M levels $\{t_i\}$, then

$\sum_{i=1}^k$

$W_k =$

$\sum_{i=1}^k \theta_1 > \theta_2 \dots \theta_k \sum_{j=1}^k P_k(\theta_1, \theta_2 | \beta) d(\theta_1, \theta_2)$

$\theta_1 < j < M$

$\theta_2 \in S_i, \theta_1 \in S_j$

as $P_k(\theta_1, \theta_2) = 0$ when $\beta(\theta_1) = \beta(\theta_2)$.

13

[form-feed in source extraction]

$\lim_{k \rightarrow \infty} \log W_k$

1

$= \lim_{k \rightarrow \infty} \log$

k

1

$= \lim_{k \rightarrow \infty} \log$

k

Z

$\sum_{i=1}^k \theta_1 > \theta_2 \dots \theta_k \sum_{j=1}^k P_k(\theta_1, \theta_2 | \beta) d(\theta_1, \theta_2)$

$\sum_{i=1}^k$

$\sum_{i=1}^k \theta_1 > \theta_2 \dots \theta_k \sum_{j=1}^k P_k(\theta_1, \theta_2 | \beta) d(\theta_1, \theta_2)$

$\theta_1 < j < M$

=

=

min

inf

(8)

$\theta_2 \in S_i, \theta_1 \in S_j$

" Z

#!

1

$= \max$

lim

log

$\sum_{i=1}^k \theta_1 > \theta_2 \dots \theta_k \sum_{j=1}^k P_k(\theta_j, \theta_i | \beta) d(\theta_1, \theta_2)$

$\theta_1 < j < M, k \rightarrow \infty$

$\theta_2 \in S_i, \theta_1 \in S_j$

[U+0013 control character]

[U+0012 control character]

[U+0003 control character]

l[U+0002 control character]

$= \max$

sup

$\log \sum_{i=1}^k \theta_1 > \theta_2 \dots \theta_k \sum_{j=1}^k P_k(\theta_j, \theta_i | \beta)$

lim

$\theta_1 < j < M, \theta_1 \in S_j, \theta_2 \in S_i, k \rightarrow \infty$

[U+0012 control character]

[U+0013 control character]

1

$= \max$

sup

$\lim \log \sum_{i=1}^k P_k(\theta_j, \theta_i | \beta)$

$\theta_1 < j < M, \theta_1 \in S_j, \theta_2 \in S_i, k \rightarrow \infty$

[U+0012 control character]

[U+0013 control character]

1

$= \lim \log \sum_{i=1}^k P_k(\theta_j, \theta_i | \beta)$

$= \min$

inf

$0 \leq i < j < M$ ϵS_j , ϵS_i
 $k \rightarrow \infty$ k
 $=$

(7)

$\theta_1 > \theta_2$

$\inf \{g(\theta_1)I(a|t_j) + g(\theta_2)I(a|t_i)\}$

$0 \leq i < j < M$ ϵS_j , ϵS_i $a \in \mathbb{R}$

(9)
(10)
(11)
(12)
(13)

min

$\inf \{g_j I(a|t_j) + g_i I(a|t_i)\}$

(14)

min

$\inf \{g_{i+1} I(a|t_{i+1}) + g_i I(a|t_i)\}$

(15)

$0 \leq i < j < M$ $a \in \mathbb{R}$

$0 \leq i < M-1$ $a \in \mathbb{R}$

The last line follows from adjacent t_i , t_{i+1} dominating the rate due to monotonicity properties.

Line (10) follows from Theorem C.1.

[U+0011 control character] i

h

[U+0010 control character] P

N [U+000F control character]

[U+000F control character]

$= \max N$

Line (9) follows from: $\forall a$ [U+000F control character] $i \geq 0$, $\limsup$ [U+000F control character] $\rightarrow 0$ [U+000F control character] $\log$

a i $\limsup$ [U+000F control character] $\rightarrow 0$ [U+000F control character] $\log(a_i)$, which

i

i

is a finite case version (with fewer assumptions) of Theorem C.1. See, e.g., Lemma 1.2.15 in (Dembo and Zeitouni, 2010) for a proof of this property.

Expanded PaperInterface specification

$\forall \{ \Omega : \text{Type } u_1 \} \{ \text{Component} : \text{Type } u_2 \} [\text{inst} : \text{MeasurableSpace } \Omega] [\text{inst}_1 : \text{Fintype Component}] [\text{DecidableEq Component}]$
 $[\text{Nonempty Component}] (\mu : \text{MeasureTheory.Measure } \Omega) [\text{MeasureTheory.IsFiniteMeasure } \mu]$
 $(P : \text{EconCSLib.Probability.FiniteMeasurableSetPartition } \mu \text{ Component}) (\text{weight} : \Omega \rightarrow \mathbb{R}) (\text{kernel} : \mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R})$
 $(\text{phi} : \text{Component} \rightarrow \Omega \rightarrow \mathbb{R}) (\text{rate } W : \text{Component} \rightarrow \mathbb{R}),$
 $(\forall (k : \mathbb{N}) (\text{component} : \text{Component}),$
 $\text{MeasureTheory.IntegrableOn } (\text{fun } x \Rightarrow \text{weight } x * \text{kernel } k \ x) (P.\text{pieceSet } \text{component}) \ \mu) \rightarrow$
 $(\forall (\text{component} : \text{Component}), \text{MeasureTheory.IntegrableOn } \text{weight } (P.\text{pieceSet } \text{component}) \ \mu) \rightarrow$
 $(\forall (\text{component} : \text{Component}), \ 0 < W \ \text{component}) \rightarrow$
 $(\forall (\text{component} : \text{Component}), \ \forall^\mu (x : \Omega) \ \partial \mu.\text{restrict } (P.\text{pieceSet } \text{component}), \ 0 \leq \text{weight } x) \rightarrow$
 $(\forall (\text{component} : \text{Component}), \ \forall^\mu (x : \Omega) \ \partial \mu.\text{restrict } (P.\text{pieceSet } \text{component}), \ \text{weight } x \leq W \ \text{component}) \rightarrow$
 $(\forall (\text{component} : \text{Component}),$
 $\text{EconCSLib.Probability.HasAEEssentialInfimum } (\mu.\text{restrict } (P.\text{pieceSet } \text{component})) (\text{phi } \text{component})$
 $(\text{rate } \text{component})) \rightarrow$
 $(\forall (\text{component} : \text{Component}),$
 $\text{EconCSLib.Probability.HasPositiveWeightNearAEEssentialInfimum } (\mu.\text{restrict } (P.\text{pieceSet } \text{component}))$
 $\text{weight } (\text{phi } \text{component}) (\text{rate } \text{component})) \rightarrow$
 $(\forall (k : \mathbb{N}) (x : \Omega), \ 0 < \text{kernel } k \ x) \rightarrow$
 $(\forall (\text{component} : \text{Component}),$
 $\forall \epsilon > 0,$
 $\forall^\epsilon (k : \mathbb{N}) \text{ in } \text{Filter.atTop}, \ \forall (x : \Omega), \ |\text{Real.log } (\text{kernel } k \ x) / \uparrow k - \text{phi } \text{component } x| \leq \epsilon) \rightarrow$
 $\exists \text{ minComponent},$
 $(\forall (\text{component} : \text{Component}), \ \text{rate } \text{minComponent} \leq \text{rate } \text{component}) \wedge$
 $\text{EconCSLib.Probability.HasExponentialRate}$
 $(\text{fun } k \Rightarrow \int (x : \Omega) \text{ in } P.\text{support}, \ \text{weight } x * \text{kernel } k \ x \ \partial \mu) (\text{rate } \text{minComponent})$

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.paper_lemmaC3_partition_integral_hasExponentialRate_of_min_component_uniform_logRate

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 42

Source locator: cited publication, lines 2366-2399

Verbatim source input

Proof. $\Rightarrow$ follows directly from Lemma C.3: $\inf_{a \in \mathbb{R}} \{g_{i+1} I(a|t_{i+1}) + g_i I(a|t_i)\} > 0$ when $t_i \neq t_{i+1}$, which holds when β is piece-wise constant with the appropriate number of levels.
 $\Leftarrow$ Consider β that is not piece-wise constant. Recall that we further assume that β is nondecreasing, and discontinuous only on a measure 0 set. Following algebra steps similar to those in Lemma C.3, but for general β :

```
Z
1
1
log W k = - lim log
w(θ1, θ2) P k (θ1, θ2 | β) d(θ1, θ2)
k → ∞ k
k → ∞ k
θ1 > θ2
[U+0012 control character]
[U+0013 control character]
1
= inf - lim log P k (θj, θi | β)
θ1 > θ2
k → ∞ k
= 0

- lim

(16)
(17)
(18)
```

Where the last line follows from β continuous at some θ_1 , and so $\lim_{\theta_2 \rightarrow \theta_1} P k (\theta_1, \theta_2 | \beta) = 1$. Intuitively, what goes wrong with continuous β is that $P k (\theta_1, \theta_2 | \beta)$ does not converge uniformly: $\forall [U+000F control character], k, \exists \theta_2 \neq \theta_1 P k (\theta_1, \theta_2) > [U+000F control character]$
14

[form-feed in source extraction]
i.e. close by items are very hard to distinguish from one another. Then, because the large deviations rate of $W k$ is dominated by the worst rates under the integral, we don't get a positive rate.

Expanded PaperInterface specification

```
∀ (μ : MeasureTheory.Measure ℝ) [MeasureTheory.SFinite μ] [MeasureTheory.IsFiniteMeasure (μ.prod μ)]
  [(μ.prod μ).IsOpenPosMeasure] {m : ℕ},
  0 < m →
  ∀ (cut : ℕ → ℝ) (hmono : Monotone cut),
    (∀ i < m + 2, cut i < cut (i + 1)) →
    ∀ (levels sampleRate : Fin (m + 2) → ℝ)
      (hlevels : GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector levels),
      (∀ (idx : Fin (m + 2)), 0 < sampleRate idx) →
      (∀ {a b : Fin (m + 2)}, ↑a ≤ ↑b → sampleRate a ≤ sampleRate b) →
      ∀ (weight : ℝ × ℝ → ℝ),
        (∀
          (component :
            { piece // GJ190ptimalBinaryRatingSystems.theorem31OrderedNontrivialPairSelected piece },
            MeasureTheory.IntegrableOn weight
              ((GJ190ptimalBinaryRatingSystems.theorem31_ordered_quality_pair_partition μ (m + 2) cut hmono
                GJ190ptimalBinaryRatingSystems.theorem31OrderedNontrivialPairSelected).pieceSet
                component)
              (μ.prod μ)) →
            (∀
              (component :
                { piece // GJ190ptimalBinaryRatingSystems.theorem31OrderedNontrivialPairSelected piece },
                ℝ ×
                  ℝ) ∂(μ.prod μ).restrict
                  ((GJ190ptimalBinaryRatingSystems.theorem31_ordered_quality_pair_partition μ (m + 2) cut hmono
                    GJ190ptimalBinaryRatingSystems.theorem31OrderedNontrivialPairSelected).pieceSet
                    component),
                  0 ≤ weight x) →
                Continuous weight →
                (∀ (component : GJ190ptimalBinaryRatingSystems.theorem31OrderedNontrivialPairComponent m),
                  0 <
                    weight
                      (GJ190ptimalBinaryRatingSystems.theorem31_ordered_quality_pair_component_midpoint cut
                        component)) →
                ∀ (lo hi : ℝ)
                  (R : GJ190ptimalBinaryRatingSystems.assumption_lemmaC4_positive_support_interval_model μ),
                  ¬GJ190ptimalBinaryRatingSystems.lemmaC4FiniteStepOnIoo R.successProb R.lo R.hi →
                  (GJ190ptimalBinaryRatingSystems.lemmaC4FiniteRangeOnIoo
                    (GJ190ptimalBinaryRatingSystems.cutpointStepSuccessProb cut levels) lo hi ∧
                      ∃ rate,
                        0 < rate ∧
                          EconCSLib.Probability.ExponentialRateCertificate
                            (GJ190ptimalBinaryRatingSystems.lemmaC4TieErasedSourceWbar μ
                              (GJ190ptimalBinaryRatingSystems.theorem31SelectedPullbackSourceWeight weight)
                              (GJ190ptimalBinaryRatingSystems.theorem31SelectedPullbackSourceKernel μ cut
                                hmono sampleRate levels hlevels))
                              rate) ∧
                            EconCSLib.Probability.HasExponentialRate
                              (GJ190ptimalBinaryRatingSystems.lemmaC4TieErasedSourceWbar μ R.weight
```



```

      (GJ190ptimalBinaryRatingSystems.lemmaC4RawSourcePbarKernel R))
    0 ^
  ∀ (rate : ℝ),
    0 < rate →
      ¬EconCSLib.Probability.ExponentialRateCertificate
        (GJ190ptimalBinaryRatingSystems.lemmaC4TieErasedSourceWbar μ R.weight
          (GJ190ptimalBinaryRatingSystems.lemmaC4RawSourcePbarKernel R))
    rate

```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.paper_lemmaC4_finite_selected_pullback_positive_rate_and_nonfiniteStep_source_zero_J
 ↪ rate

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 43

Source locator: cited publication, lines 2405-2409

Verbatim source input

convex, with minima at $a = b$, when $a, b \in (0, 1)$. Note that $\inf a \{g_i \text{KL}(a||t_i) + g(i+1)\text{KL}(a||t_{i+1})\}$, for all feasible g , is also continuous and strictly convex in t_i, t_{i+1} , with minima at $t_i = t_{i+1}$. One consequence of the above fact is that fixing either t_i or t_{i+1} and moving the other farther away monotonically increases KL, while moving it closer decreases KL.

C.4.1

Expanded PaperInterface specification

```
∀ {gHi gLo pLo pHi : ℝ},
  0 < gHi →
    0 < gLo →
      0 < pLo →
        pLo < pHi →
          pHi < 1 →
            ContinuousAt (fun q => EconCSLib.Probability.weightedBernoulliClosedThresholdRate gHi gLo q.1 q.2)
              (pHi, pLo) ∧
              (∀ p ∈ Set.Ioo 0 1, EconCSLib.Probability.weightedBernoulliClosedThresholdRate gHi gLo p p = 0) ∧
              0 < EconCSLib.Probability.weightedBernoulliClosedThresholdRate gHi gLo pHi pLo ∧
              StrictMonoOn (fun x => EconCSLib.Probability.weightedBernoulliClosedThresholdRate gHi gLo x pLo)
                (Set.Icc pLo pHi) ∧
              StrictAntiOn (fun x => EconCSLib.Probability.weightedBernoulliClosedThresholdRate gHi gLo pHi x)
                (Set.Icc pLo pHi)
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_remarkC2_weighted_kl_separation_monotonicity

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 44

Source locator: cited publication, lines 2670-2714

Verbatim source input

Additional necessary lemmas

Now, we begin the set-up that will lead to a proof for Theorem 3.2. It turns out that proving the theorem requires, in the process, essentially proving our convergence result with $M \rightarrow \infty$, Theorem B.1. For Theorem 3.2, we need a $\hookrightarrow$ lower bound for t_1 as a function of M . This seems hard to do in general. Luckily, in our case, there is a property for how t_* changes when M is doubled. Using this property, we can derive that $t_* \geq 0(M^{-3})$.

Recall that step-wise increasing β with M intervals $S_i = [s_i, s_{i+1})$ has levels $\{t_i\}_M$ $i=0$, where $t_0 = 1$, $t_{M-1} = 1$, and $s_0 = 0$, $s_M = 1$. Furthermore, we use the following notation for the large deviation rate

```
#
"
g
g

ri = -(gi-1 + gi) log (1 - ti-1)

gi-1
gi-1 +gi

(1 - ti)

gi
gi-1 +gi

i-1
gi-1 +gi

+ ti-1

g

i
+gi

ti i-1

(23)

for i ∈ {1 . . . M - 1}, which implies r1 = -g1 log(1 - t1) and rM-1 = -gM-2 log(tM-2).
We further use rM-1 to be the rate achieved by the optimal  $\beta_M$  with  $M$  intervals.
-1
Lemma C.5. Suppose  $g$  uniform, i.e.  $g_i = 1$ ,  $\forall i$  and that  $\beta_M$  has values  $\{t_i\}_M$ 
```

Expanded PaperInterface specification

```
∀ {m : ℕ} (successProb sampleRate : Fin (m + 2) → ℝ) (i : Fin (m + 1)),
GJ190OptimalBinaryRatingSystems.binaryEndpointAwareAdjacentRate successProb sampleRate i =
  if i = 0 then
    sampleRate (GJ190OptimalBinaryRatingSystems.adjacentHighIndex i) *
    -Real.log (1 - successProb (GJ190OptimalBinaryRatingSystems.adjacentHighIndex i))
  else
    if i = m then
      sampleRate (GJ190OptimalBinaryRatingSystems.adjacentLowIndex i) *
      -Real.log (successProb (GJ190OptimalBinaryRatingSystems.adjacentLowIndex i))
    else
      EconCSLib.Probability.weightedBernoulliClosedThresholdRate
        (sampleRate (GJ190OptimalBinaryRatingSystems.adjacentHighIndex i))
        (sampleRate (GJ190OptimalBinaryRatingSystems.adjacentLowIndex i))
        (successProb (GJ190OptimalBinaryRatingSystems.adjacentHighIndex i))
        (successProb (GJ190OptimalBinaryRatingSystems.adjacentLowIndex i))
```

Lean proof endpoint (built separately)

GJ190OptimalBinaryRatingSystems.PaperInterface.source_appendixC5_rate_notation_eq23

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 45

Source locator: cited publication, lines 2715-2895

Verbatim source input

```
i=0[U+0001 control character] . Then  $\beta_{2M-1}^{2M-2}$ 
 $\beta_{2M-2}^{2M-1}$ 
1
0
0
0
has values  $\{t_i\}_{i=0}^{2M-1}$ , where  $t_{2i} = t_i$ ,  $\forall i \in \{0, \dots, M-1\}$ ,  $t_1 = 2t_0 - t_{-1}$  and  $t_{2M-3} =$ 
 $\sqrt{\frac{1}{2(1+t_{M-2})}}$ .

Proof. We first set the values  $t_{2i} = t_i$  and then optimally choose the remaining values  $t_{0k}$ ,  $k$  odd.
Then, we show that the resulting large deviation rates between all adjacent pairs are equal. Then,
by the proof of Lemma 3.1, which showed that equalizing the rates between adjacent intervals is a
sufficient condition for optimality,  $\beta_{2M-1}^{2M-2}$  has the levels  $\{t_{0i}\}_{i=0}^{2M-1}$ .
i=0 .
Let  $r_0$  denote rates between adjacent  $t_0$  as  $r$  does for  $t$ . Supposing  $t_2 = t_1$ , we find  $t_1$  such that
 $r_1 = r_2$  and  $t_1 < t_2$ .
[U+0014 control character]q
[U+0015 control character]
q
0
0
0
0
0
0
-  $\log(1 - t_1) = -2 \log$ 
 $(1 - t_1)(1 - t_2) + t_1 t_2$ 
q
 $\Rightarrow 1 - t_1 = (1 - t_1)(1 - t_2) + t_1 t_2 + 2(1 - t_1)(1 - t_2)t_1 t_2$ 
[U+0012 control character]q
[U+0013 control character]
q
[U+0001 control character]
 $\sqrt{\frac{1}{2(1+t_{M-2})}}$ 
1
1
0
0
 $\Rightarrow t_1 =$ 
 $\frac{1 - t_2}{1 - t_1}$ 
2
2
1
0
0
0
Similarly,  $r_{2M-3} = r_{2M-2}$  when  $t_{2M-3} = 2(1 +$ 
 $t_{2M-4})$  by choosing such  $t_{01}$ ,  $t_{2M-3}$ .

 $\sqrt{\frac{1}{2(1+t_{M-2})}}$ . It follows that  $r_{10} = r_{20} = r_{2M-3}$ .

17

[form-feed in source extraction]
0
Next, we find  $t_{0k} \in (t_{0k-1}, t_{0k+1})$  for  $k \in \{3, 5, \dots, 2M-5\}$  such that the rates  $r_{k0} = r_{k+1}$ .
.
q
q
hq
i
hq
i
0
0
0
0
0
0
 $(1 - t_k)(1 - t_{k-1}) + t_k t_{k-1} = -2 \log$ 
 $(1 - t_k)(1 - t_{k+1}) + t_{0k} t_{0k+1}$ 
-2  $\log$ 
q
[U+F8F9 private-use glyph]2
[U+F8EE private-use glyph]q
0
0
1
```



```

∀ {m : ℕ},
  0 < m →
  ∀ {oldLevels : Fin (m + 2) → ℝ},
    GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector oldLevels →
      (GJ190ptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize oldLevels fun x => 1) →
        GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector
          (GJ190ptimalBinaryRatingSystems.uniformDoubledEndpointLevels oldLevels) ∧
          (GJ190ptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize
            (GJ190ptimalBinaryRatingSystems.uniformDoubledEndpointLevels oldLevels) fun x => 1) ∧
            EconCSLib.Optimization.IsMaximizerOn GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector
              (fun xs => GJ190ptimalBinaryRatingSystems.binaryEndpointAwareAdjacentRate0bjective xs fun x => 1)
              (GJ190ptimalBinaryRatingSystems.uniformDoubledEndpointLevels oldLevels)

```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_lemmaC5_uniform_doubled_chain

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 46

Source locator: cited publication, lines 2884-3038

Verbatim source input

```
[U+0014 control character]√
√
1 - ti - 1 - ti-1
√
where c =
√
ti-1 - ti

(24)
(25)

The proof for (25) is algebraically tedious and is shown in Remark C.3 below.
Then, by the proof of Lemma 3.1, which shows that equalizing the rates inside the minimization
-2
terms implies an optimal {ti }, β2M -1 has the levels {t0i }2M
i=0 .
Remark C.3.
"s[U+0012 control character]

c
1-
1+c

r

[U+0013 control character]
(1 - ti ) +

#2

i
p
p
lh
1 + (1 - ti-1 )(1 - ti ) + ti-1 ti
2
[U+0015 control character]2
[U+0014 control character]√
√
1 - ti - 1 - ti-1
√
where c =
√
ti-1 - ti

c
ti
1+c

=

√
√
√
√
Proof. Let x = ti , y = 1 - ti , z = ti-1 , and w = 1 - ti-1 . Note that x > z, w > y, y =
1 - x2 , w = 1 - z 2 . Then,
c
(y - w)2
1
(x - z)2
=
, and
=
c+1
2 - 2xz - 2yw
c+1
2 - 2xz - 2yw
1
(To show the above two equalities, factor out (x-z)
2
from numerator and denominator, and substi2
2
tute y = 1 - x , w = 1 - z ).

18

[form-feed in source extraction]
Now, the left hand side:
"s[U+0012 control character]

c
1-
1+c

#2
c
(1 - ti ) +
ti
```

```

1+c
i2
hp
p
(x - z)2 y 2 + (y - w)2 x2
r

[U+0013 control character]

1
2 - 2xz - 2yw
(x - z)2 y 2 + (y - w)2 x2 + 2xy(x - z)(w - y)
=
2 - 2xz - 2yw
2
2
2
2
z y + w x - 2wxyz
=
2 - 2xz - 2yw
=

p
p
(y - w)2 = w - y, (x - z)2 = x - z

The right hand side:
i
p
p
lh
1 + (1 - ti-1)(1 - ti) + ti-1 ti
2
1
= [1 + (wy + xz)]
2
Multiplying both sides by 2 - 2xz - 2yw, we have:
"s[U+0012 control character]

```

```

c
1-
1+c

r

[U+0013 control character]
(1 - ti) +

c
ti
1+c

#2
=

i
p
p
lh
1 + (1 - ti-1)(1 - ti) + ti-1 ti
2

≡ z 2 y 2 + w2 x2 - 2wxyz = 1 - (wy + xz)2
≡ z 2 (1 - x2) + (1 - z 2)x2 - 2wxyz = 1 - w2 y 2 - x2 z 2 - 2wxyz
≡ z 2 - 2x2 z 2 + x2 = 1 - (1 - z 2)(1 - x2) - x2 z 2
≡ 0=0

```

Expanded PaperInterface specification

```

(∀ {m k : ℕ},
  0 < k →
    ∀ (hkm : k < m) (oldLevels : Fin (m + 2) → ℝ),
      GJ190ptimalBinaryRatingSystems.uniformDoubledEndpointLevels oldLevels (2 * k + 1, ∞) =
        EconCSLib.Probability.bernoulliInteriorEqualSplit (oldLevels (k, ∞)) (oldLevels (k + 1, ∞))) ∧
  ∀ {pLo pHi : ℝ},
    0 ≤ pLo →
      pHi ≤ 1 →
        pLo < pHi →
          EconCSLib.Probability.weightedBernoulliClosedThresholdRate 1 1
            (EconCSLib.Probability.bernoulliInteriorEqualSplit pLo pHi) pLo =
            EconCSLib.Probability.weightedBernoulliClosedThresholdRate 1 1 pHi
            (EconCSLib.Probability.bernoulliInteriorEqualSplit pLo pHi)

```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_lemmaC5_refinement_equations24_25

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 47

Source locator: cited publication, lines 3042-3092

Verbatim source input

0

Proof. Let $M = 2N$, $M \theta = 2N + 1 - 1$, for some N . We show that $rM \leq 12 rM$. The corollary follows
0
by noting that $rK < rK \forall K \theta > K$ and that $rK < \infty$, $\forall K$.
0

0

M
 $rM - rM = -\log tM$
 $M - 2 + \log tM \theta - 2$
[U+0014 control character]
[U+0015 control character]
q
1 1 M
M
 $= -\log tM - 2 + \log$
+
t
2 2 M -2
[U+F8EE private-use glyph]
[U+F8F9 private-use glyph]
1 1
1
1 [U+F8FB private-use glyph]
 $= \log [U+F8F0 private-use glyph] M + q$
2 tM -2 2 tM

Lemma C.5

M -2

1
 $\geq -\log tM$
M -2
2
1
0
 $\Rightarrow rM \leq rM$
2

q

19

M
tM
M -2 $\geq tM - 2$

[form-feed in source extraction]
M

Expanded PaperInterface specification

```
∀ (levels : (N : ℕ) → Fin (N + 1 + 2) → ℝ),
  (∀ (N : ℕ), GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector (levels N)) →
  (∀ (N : ℕ), GJ190ptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize (levels N) fun x => 1) →
  Filter.Tendsto
    (fun N =>
      GJ190ptimalBinaryRatingSystems.binaryEndpointAwareAdjacentRate (levels N) (fun x => 1)
      GJ190ptimalBinaryRatingSystems.lastAdjacentIndex)
    Filter.atTop (nhds 0)
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.paper_corollaryC2_uniform_equalized_last_rate_tendsto_zero

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 48

Source locator: cited publication, lines 3094-3125

Verbatim source input

$k - tk - 1 < \delta$.

Proof. This corollary follows directly from Corollary C.1. If the rates are upper bounded, then so are the level differences.

We first find where the rate is minimized given a width between levels of δ

$$x_m = \arg \min_x -2 \log \frac{(1 - x - \delta)(1 - x) + x(x + \delta)}{x}$$

$$\frac{1}{2} = -\frac{1}{2} \log \frac{(1 - x - \delta)(1 - x) + x(x + \delta)}{x}$$

Then given an upper bound of $[U+000F \text{ control character}]$ on the rate, there is a bound on δ determined by the largest possible difference at levels symmetric around 2^1 .

$$\frac{1}{2} = -\log \frac{(1 - \delta)(1 + \delta)}{1 - 4\delta^2}$$
$$\frac{1}{2} \geq [U+000F \text{ control character}] \text{ when } \delta > \frac{1 - e^{-[U+000F \text{ control character}]}}{2}$$

Expanded PaperInterface specification

$$\forall (\text{levels} : (N : \mathbb{N}) \rightarrow \text{Fin } (N + 1 + 2) \rightarrow \mathbb{R}),$$
$$(\forall (N : \mathbb{N}), \text{GJ190OptimalBinaryRatingSystems.BinaryEndpointLevelVector } (\text{levels } N)) \rightarrow$$
$$(\forall (N : \mathbb{N}), \text{GJ190OptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize } (\text{levels } N) \text{ fun } x \Rightarrow 1) \rightarrow$$
$$\text{Filter.Tendsto } (\text{fun } N \Rightarrow \text{GJ190OptimalBinaryRatingSystems.binaryEndpointAdjacentMaxWidth } (\text{levels } N)) \text{ Filter.atTop } (nhds \ 0)$$

Lean proof endpoint (built separately)

`GJ190OptimalBinaryRatingSystems.PaperInterface.paper_corollaryC2_uniform_equalized_adjacent_mesh_tendsto_zero`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 49

Source locator: cited publication, lines 3131-3159

Verbatim source input

Proof. Note that, with uniform matching, $\forall x \in (0, 1], y \in [0, 1 - x]$ the rate with values $t_{i-1} = y$, $t_i = y + x$ is no more than the last with $t_{M-2} = 1 - x$. With width x , in other words, the extreme points have a larger rates than the middle points. For $i \in \{1, M - 1\}$:

$$r_i = \inf_a \{g_{i-1} KL(a || t_{i-1}) + g_i KL(a || t_i)\}$$
$$= \inf_a \{KL(a || y) + KL(a || y + x)\}$$

uniform matching

$$= -2 \log \frac{(1 - y)^2 (1 - y - x)^2 + y^2 (y + x)^2}{(26)}$$
$$= -\log \frac{(1 - y)(1 - y - x) + y(y + x) + 2 [(1 - y)(1 - y - x)y(y + x)]^{1/2}}{(26)}$$

Where line (26) follows from line (22).

By the proof of Lemma 3.1, the optimal levels equalize the rates between each level. Then, when g is non-decreasing, $g_{M-2} \geq g_i$, $\forall i \in \{1 \dots M - 3\}$. Then, at the same level differences, the rate corresponding to the last level is no smaller. Thus, to equalize the rates, the last width must be no larger than any other width. Thus, $t_{M-2} \geq 1 - L_1$.

$N + 1 - 1$

Expanded PaperInterface specification

```
∀ {m : ℕ},
  0 < m →
  ∀ {levels sampleRate : Fin (m + 2) → ℝ},
    GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector levels →
      GJ190ptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize levels sampleRate →
        (∀ (idx : Fin (m + 2)), 0 < sampleRate idx) →
          (∀ {a b : Fin (m + 2)}, ↑a ≤ ↑b → sampleRate a ≤ sampleRate b) →
            1 - 1 / ↑(m + 1) ≤
              levels
                (GJ190ptimalBinaryRatingSystems.adjacentLowIndex GJ190ptimalBinaryRatingSystems.lastAdjacentIndex)
```

Lean proof endpoint (built separately)

`GJ190ptimalBinaryRatingSystems.PaperInterface.lemmaC6_monotone_matching_penultimate_level_bound`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 50

Source locator: cited publication, lines 3161-3233

Verbatim source input

20

N

$\geq 15 \ r^2 \ .$

[form-feed in source extraction]

1

Proof. Let $K = 2N$, $K \ 0 = 2N + 1 - 1$. Note that tK
 $K-1 \geq 2$ by Lemma C.6.

0

0

K

$rK - rK = - \log \ tK$

$K-2 + \log \ tK \ 0 -2$

[U+0014 control character]

[U+0015 control character]

q

1 1 K

K

$= - \log \ tK-2 + \log$

+

t

2 2 K-2

[U+F8EE private-use glyph]

[U+F8F9 private-use glyph]

1

1

1

1

[U+F8FB private-use glyph]

$= \log \ [U+F8F0 \text{ private-use glyph}] \ K + q$

2 $tK-2 \ 2 \ tK$

Lemma C.5

$K-2$

$\leq \log$

tK

$K-2$

[U+0001 control character]

[U+0014 control character]

[U+0015 control character]

[U+0001 control character]- 45

1 K [U+0001 control character]-1 1 K [U+0001 control character]- 21

1

K

K

$\leq tK-2$

when $tK-2 \in$

t

+

t

,1

2 K-2

2 K-2

2

- 54

1

0

$\Rightarrow rK \geq rK$

5

-3 .

Lemma C.8. With uniform matching ($g_i = 1$), $\exists C > 0$ s.t. $\forall M, tM$

Expanded PaperInterface specification

$\forall \{m : \mathbb{N}\},$

$0 < m \rightarrow$

$\forall \{\text{oldLevels} : \text{Fin } (m + 2) \rightarrow \mathbb{R}\},$

$\text{GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector oldLevels} \rightarrow$

$(\text{GJ190ptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize oldLevels fun } x \Rightarrow 1) \rightarrow$

$(1 / 5 * \text{GJ190ptimalBinaryRatingSystems.binaryEndpointAwareAdjacentRateObjective oldLevels fun } x \Rightarrow 1) \leq$

$\text{GJ190ptimalBinaryRatingSystems.binaryEndpointAwareAdjacentRateObjective}$

$(\text{GJ190ptimalBinaryRatingSystems.lemmaC5_uniform_doubled_endpoint_levels oldLevels}) \text{ fun } x \Rightarrow 1$

Lean proof endpoint (built separately)

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

Review row 51

Source locator: cited publication, lines 3234-3267

Verbatim source input

$1 \geq CM$

Proof. By Lemma C.7, $\exists C_2 > 0$ s.t. $rM \geq C_2 5^{-d \log_2 M} e$. Then

$$\begin{aligned} & \frac{1}{M} \log(1 - tM) \\ & \geq C_2 5^{-d \log_2 M} e \\ & \Rightarrow tM \\ & 1 \geq 1 - \exp -C_2 5 \\ & \Rightarrow tM \\ & \geq 1 - \exp -C_3 M \log_5 2 \\ & \Rightarrow tM \\ & \geq 1 - \exp -C_3 M \log_5 2 \\ & \Rightarrow \exists C > 0 \text{ s.t. } t_1 \geq CM^{-3} \end{aligned}$$

$e^{-x} \leq 1 -$

$\geq$

e^{-1}
x for $x \in [0, 1]$
e

-3 .

Corollary C.3. With monotonically non-decreasing g , $\exists C > 0$ s.t. $\forall M, tM$

Expanded PaperInterface specification

```
∀ {m : ℕ},
  0 < m →
    ∀ {levels : Fin (m + 2) → ℝ},
      GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector levels →
        (GJ190ptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize levels fun x => 1) →
          (1 / ↑(m + 1)) ^ 2 / 2 ≤
            levels (GJ190ptimalBinaryRatingSystems.adjacentHighIndex GJ190ptimalBinaryRatingSystems.firstAdjacentIndex)
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_lemmaC8_uniform_first_level_polynomial_lower_bound

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 52

Source locator: cited publication, lines 3268-3274

Verbatim source input

$1 \geq CM$

Proof. The result follows from noting that tM with uniform matching lower bounds the first value with any other monotonically non-decreasing g , which is a direct application of Lemma B.1 – scale g such that $g_1 = 1$. Then, $g_j \geq 1$, $j > 1$ and $g_0 \leq 1$. Then, the condition of the lemma holds. Lemma C.9. The run-time of NestedBisection is $O(M \log^2 1/\delta)$, where δ is the bisection grid width

Expanded PaperInterface specification

```

∀ {m : ℕ},
  0 < m →
    ∀ {levels sampleRate : Fin (m + 2) → ℝ},
      GJ190ptimalBinaryRatingSystems.BinaryEndpointLevelVector levels →
        GJ190ptimalBinaryRatingSystems.BinaryEndpointAwareAdjacentRatesEqualize levels sampleRate →
          (∀ (idx : Fin (m + 2)), 0 < sampleRate idx) →
            (∀ {a b : Fin (m + 2)}, ↑a ≤ ↑b → sampleRate a ≤ sampleRate b) →
              sampleRate
                (GJ190ptimalBinaryRatingSystems.adjacentHighIndex
                  GJ190ptimalBinaryRatingSystems.firstAdjacentIndex) =
                  1 →
                    (1 / ↑(m + 1)) ^ 2 / 2 ≤
                      levels
                        (GJ190ptimalBinaryRatingSystems.adjacentHighIndex GJ190ptimalBinaryRatingSystems.firstAdjacentIndex)
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.corollaryC3_monotone_scaled_first_level_ge_half_inv_adjacent_count_sq

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 53

Source locator: cited publication, lines 3275-3279

Verbatim source input

and M is the number of intervals.
Proof. The outer bisection, in main, runs at most $\log_2 26 + 1$ iterations. Each outer iteration calls BisectNextLevel M - 3 times, and the inner bisection in each call runs for at most $\log_2 26$ iterations. Thus the run-time of algorithm is $O(M \log_2 16)$.

Expanded PaperInterface specification

```

∀ {M : ℕ},
  0 < M →
    ∀ {delta : ℝ},
      0 < delta →
        ∃ L,
          1 ≤ delta * 2 ^ L ∧
            ↑(L + 1) ≤ Real.logb 2 (max 1 (1 / delta)) + 2 ∧
              ↑(GJ190ptimalBinaryRatingSystems.nestedBisection0operationCount M (L + 1) L) ≤
                ↑M * (Real.logb 2 (max 1 (1 / delta)) + 2) ^ 2

```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_lemmaC9_nested_bisection_runtime_log_squared

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 54

Source locator: cited publication, lines 3467-3589

Verbatim source input

Proof of Theorem B.1

w denote the optimal β with M intervals for weight function w, with intervals sw_M and levels

Let β_M

tw_M . Let $qw_M(\theta) = i/M$ when $\theta \in [sw_M$

, sw_M

i

$i+1)$, i.e. the quantile of interval item of type θ is in.

Then we have the following convergence result for β_M .

Theorem B.1. Let g be uniform. Suppose w such that qw_M converges uniformly. Then, $\forall C \in$

w uniformly as $N \rightarrow \infty$.

w

$N, \exists \beta$ w s.t. βC^2

$N+1 \rightarrow \beta$

hProof. Note that

[U+0011 control character] the condition on q implies that $\exists M$ s.t.

M

M

$sbx_{\theta} M c, sdx_{\theta} M e$.

$\forall M > M, \forall \theta, \exists x_{\theta}$ such that $\theta \in$

h

[U+0011 control character]

M

Let $M_{\theta} = 2M - 1, M_{\theta\theta} = 4M - 3, M_q = 2q M - 2q + 1. \theta \in sM$

,

s

$\Rightarrow \beta_M(\theta) =$

$bx_{\theta} M c dx_{\theta} M e$

h

i

M

M

tM

$bx_{\theta} M c \in tbx_{\theta} M c-1, tbx_{\theta} M c+1$. Then,

θ

$\beta_M \theta(\theta) = tM$

$bx_{\theta} M_{\theta} c$

θ

$= tM$

$bx_{\theta} (2M-1)c$

h $\theta\theta$

i

M_{θ}

$\in tM$

,

t

$2bx_{\theta} M c-2 2bx_{\theta} M c+2$

i

h

M

$c tM$

,

t

$bx_{\theta} M c-1 bx_{\theta} M c+1$

23

Lemma C.5

[form-feed in source extraction]

And, for general q,

q

$\beta_M q(\theta) = tM$

$bx_{\theta} (2q M - 2q + 1)c$

i

h θq

M_q

,

t

$\in tM$

q

q

q

$bx_{\theta} (2 M) c+1$

$bx_{\theta} 2 M c-2$

h q

i

M_q

$c tM$

,

t

q

q

```

q
2 bx0 M c-2
2 bx0 M c+1
h
i
M
c tM
'
t
bx0 M c-1 bx0 M c+1

```

Lemma C.5

```

h
i
M
Then,  $\forall N \theta > 1, \theta: \beta_{2N \theta M - 2N \theta + 1}(\theta) \in tM$ 
'
t
bx0 M c-1 bx0 M c+1 and
M
 $|\beta_{2N \theta M - 2N \theta + 1}(\theta) - \beta_M(\theta)| \leq tM$ 
bx0 M c+1 - tbx0 M c-1
0
0

```

K
By Corollary C.2, $\forall \delta > 0, \exists K$ s.t. $\forall K \theta > K, tK$
 $bx0 K \theta c+1 - tbx0 K \theta c-1 < 2\delta$.
By the Cauchy criterion, $\exists \beta$ s.t. $\beta(C-1)2N + 1 \rightarrow \beta$ uniformly.
By change of variables, $\exists \beta$ s.t. $\beta_{2N} + 1 \rightarrow \beta$ uniformly.

Corollary C.4. For Kendall's tau and Spearman's rho correlation measures, $\exists \beta$ s.t. $\beta_{2N} \rightarrow \beta$

Expanded PaperInterface specification

```

 $\forall \{M \ q : \mathbb{N}\},$ 
 $\theta < M \rightarrow$ 
 $\forall \{\theta : \mathbb{R}\},$ 
 $\theta \leq \theta \rightarrow$ 
 $\theta \leq 1 \rightarrow$ 
have old :=  $\lfloor tM * \theta \rfloor$ ;;
have refined :=  $\lfloor t(GJ190ptimalBinaryRatingSystems.theoremB1SourceDoubledIndexIterate \ M \ q) * \theta \rfloor$ ;;
refined  $\leq 2^q * old + 2^q \wedge 2^q * old \leq refined + 2^q$ 

```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_theoremB1_proof_selector_nesting

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 55

Source locator: cited publication, lines 3590-3593

Verbatim source input

uniformly as $N \rightarrow \infty$.
Proof. For Kendall's tau and Spearman's rho, $\{s_i\}$ is spaced such that $\forall i, j, s_i - s_{i-1} = s_j - s_{j-1}$.
Thus, $x_\theta = \theta$ meets the criterion.

Expanded PaperInterface specification

```
∀ (C : ℕ),
  (∀ (M : ℕ),
    Nonempty (Fin M) →
      0 < M →
        EconCSLib.Optimization.IsLexicographicMaximizerOn
          (GJ190ptimalBinaryRatingSystems.ProofBridge.theorem31KendallFiniteDesignFeasible M)
          (GJ190ptimalBinaryRatingSystems.ProofBridge.theorem31KendallFiniteDesignValue M)
          (GJ190ptimalBinaryRatingSystems.ProofBridge.theorem31FiniteDesignEndpointRate M)
          (GJ190ptimalBinaryRatingSystems.ProofBridge.theorem31CanonicalUniformEndpointDesign M)) ∧
  (∀ (M : ℕ),
    Nonempty (Fin M) →
      0 < M →
        EconCSLib.Optimization.IsLexicographicMaximizerOn
          (GJ190ptimalBinaryRatingSystems.ProofBridge.theorem31SpearmanFiniteDesignFeasible M)
          (GJ190ptimalBinaryRatingSystems.ProofBridge.theorem31SpearmanFiniteDesignValue M)
          (GJ190ptimalBinaryRatingSystems.ProofBridge.theorem31FiniteDesignEndpointRate M)
          (GJ190ptimalBinaryRatingSystems.ProofBridge.theorem31CanonicalUniformEndpointDesign M)) ∧
  ∃ betaLimit,
    TendstoUniformlyOn
      (fun N θ =>
        GJ190ptimalBinaryRatingSystems.canonicalUniformEqualizedClampedFloorBetaSeq
          (GJ190ptimalBinaryRatingSystems.theoremB1SubsequenceIndex C N) θ)
    betaLimit Filter.atTop (Set.Icc 0 1)
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.source_corollaryC4_kendall_spearman_subsequence

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 56

Source locator: cited publication, lines 3595-3614

Verbatim source input

Kendall’s tau and Spearman’s rho related proofs

Definition C.1 (see e.g. Embrechts et al. (2003); Nelsen (2007)). The population version of Kendall-tau correlation between item true quality and rating scores is proportional to

$$\frac{\tau}{\sqrt{2}} \frac{\sum_{k=1}^K \sum_{\theta_1 > \theta_2} (\theta_1 - \theta_2) d_{\theta_1} d_{\theta_2}}{\sqrt{\sum_{k=1}^K \sum_{\theta_1 > \theta_2} d_{\theta_1} d_{\theta_2}}}$$

Similarly, given items with qualities $\theta_1, \theta_2, \theta_3$, the population version of Spearman’s rho correlation between item true quality and rating scores is

$$\frac{\rho}{\sqrt{6}} \frac{\sum_{k=1}^K \sum_{\theta_1 > \theta_2 > \theta_3} (\theta_1 - \theta_2) d_{\theta_1} d_{\theta_2} d_{\theta_3}}{\sqrt{\sum_{k=1}^K \sum_{\theta_1 > \theta_2 > \theta_3} d_{\theta_1} d_{\theta_2} d_{\theta_3}}}$$

Lemma C.10. Spearman’s ρ can also be written as being proportional to

Expanded PaperInterface specification

$$\forall (M : \mathbb{N}) (s : \mathbb{N} \rightarrow \mathbb{R}), \\ \text{GJ190OptimalBinaryRatingSystems.kendallConstantWeightIntervalObjective } M \ s = (1 - \sum_i (s(i+1) - s(i))^2) / 2 \wedge \\ \text{GJ190OptimalBinaryRatingSystems.spearmanLinearWeightIntervalObjective } M \ s = (1 - \sum_i (s(i+1) - s(i))^3) / 6$$

Lean proof endpoint (built separately)

GJ190OptimalBinaryRatingSystems.PaperInterface.definitionC1_kendall_spearman_population_objectives

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 57

Source locator: cited publication, lines 3615-3697

Verbatim source input

```

i.e. with  $w(\theta_1, \theta_2) = (\theta_1 - \theta_2)$ .
Proof. Recall  $P_k(\theta_1, \theta_3) =$ 
 $P(r((\theta_1 - \theta_2)(x_{k1} - x_{k3}) > 0))$ 
 $Z$ 
 $Z$ 
 $=$ 
 $P(r(x_{k1} - x_{k3} > 0)d\theta_1 d\theta_2 d\theta_3 +$ 
 $\theta_1 > \theta_2, \theta_3$ 
 $Z$ 
 $=$ 
 $P(r(x_{k1} - x_{k3} > 0)$ 
 $R$ 
 $\theta_1 > \theta_2 (\theta_1 - \theta_2) P_k(\theta_1, \theta_2) d\theta_1 d\theta_2,$ 
 $P(r(x_{k1} - x_{k3} < 0)d\theta_1 d\theta_2 d\theta_3$ 
 $\theta_1 < \theta_2, \theta_3$ 
 $[U+0014 \text{ control character}]Z \theta_1$ 
 $[U+0015 \text{ control character}]$ 
 $Z$ 
 $P(r(x_{k1} - x_{k3} < 0)$ 
 $[U+0014 \text{ control character}]Z 1$ 
 $[U+0015 \text{ control character}]$ 
 $d\theta_2 d\theta_1 d\theta_3 +$ 
 $d\theta_2 d\theta_1 d\theta_3$ 
 $\theta, \theta$ 
 $\theta_2 = \theta$ 
 $\theta_1, \theta_3$ 
 $\theta_2 = \theta_1$ 
 $Z 1 3h$ 
 $i$ 
 $=$ 
 $P(r(x_{k1} - x_{k3} > 0)\theta_1] + P(r(x_{k1} - x_{k3} < 0)(1 - \theta_1) d\theta_1 d\theta_3$ 
 $\theta, \theta$ 
 $Z 1 3h$ 
 $h$ 
 $ii$ 
 $=$ 
 $P(r(x_{k1} - x_{k3} < 0) + \theta_1 P(r(x_{k1} - x_{k3} > 0) - P(r(x_{k1} - x_{k3} < 0) d\theta_1 d\theta_3$ 
 $\theta_1, \theta_3$ 
24

```

$$\begin{aligned} & \text{Similarly,} \\ & P \int_{\theta_1}^{\theta_3} r((\theta_1 - \theta_2)(x_{k1} - x_{k3}) < 0) = \\ & P \int_{\theta_1}^{\theta_3} r(x_{k1} - x_{k3} > 0) + \theta_1 P \int_{\theta_1}^{\theta_3} r(x_{k1} - x_{k3} < 0) - P \int_{\theta_1}^{\theta_3} r(x_{k1} - x_{k3} > 0) d\theta_1 d\theta_3 \\ & = P \int_{\theta_1}^{\theta_3} r(x_{k3} - x_{k1} > 0) + \theta_3 P \int_{\theta_1}^{\theta_3} r(x_{k3} - x_{k1} < 0) - P \int_{\theta_1}^{\theta_3} r(x_{k3} - x_{k1} > 0) d\theta_1 d\theta_3 \end{aligned}$$

Where the second equality follows from θ_1, θ_3 interchangeable. Then

$$\begin{aligned} & \int_{\theta_1}^{\theta_2} \int_{\theta_1}^{\theta_2} (1 - \theta_1 - \theta_2) P_k(\theta_1, \theta_2) d\theta_1 d\theta_2 \\ & = \int_{\theta_1}^{\theta_2} \int_{\theta_1}^{\theta_2} (1 - \theta_1 - \theta_2) P_k(\theta_1, \theta_2) d\theta_1 d\theta_2 \\ & = \int_{\theta_1}^{\theta_2} \int_{\theta_1}^{\theta_2} (1 - \theta_1 - \theta_2) P_k(\theta_1, \theta_2) d\theta_1 d\theta_2 \end{aligned}$$

Note that Spearman's ρ is similar to Kendall's τ with an additional weighting for how far apart the two values that are flipped are.

Lemma C.11. When w is constant, i.e. for Kendall's τ rank correlation, the intervals s that

Expanded PaperInterface specification

```

 $\forall (M : \mathbb{N}) (s : \mathbb{N} \rightarrow \mathbb{R}),$ 
   $s\ 0 = 0 \rightarrow$ 
   $s\ M = 1 \rightarrow$ 
    GJ190ptimalBinaryRatingSystems.spearmanLinearWeightOrderedPairIntervalObjective M s =
      GJ190ptimalBinaryRatingSystems.spearmanLinearWeightIntervalObjective M s

```

Lean proof endpoint (built separately)

```

GJ190ptimalBinaryRatingSystems.PaperInterface.paper_lemmaC10_spearman_linear_weight_ordered_pair_interval_objective_eq_gap_objective
↔ ive

```

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 58

Source locator: cited publication, lines 3698-3712

Verbatim source input

maximize (20),
X
X Z
w(θ1 , θ2)d(θ1 , θ2) =
(s_{i+1} - s_i)(s_{j+1} - s_j)
(27)
0 ≤ i < j < M

θ2 ∈ s_i , θ1 ∈ s_j

0 ≤ i < j < M

, are {s_i = M_i }_M
i=0 .
Lemma C.12. When w is (θ1 - θ2), i.e. for Spearman’s ρ rank correlation, the intervals s that

Expanded PaperInterface specification

∀ {M : ℕ} [Nonempty (Fin M)] (s : ℕ → ℝ),
s 0 = 0 → s M = 1 → (∑ i, ∑ j, if i < j then (s (↑i + 1) - s ↑i) * (s (↑j + 1) - s ↑j) else 0) ≤ (1 - (↑M)⁻¹) / 2

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.paper_lemmaC11_kendall_constant_weight_ordered_pair_interval_objective_le_equispaced

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 59

Source locator: cited publication, lines 3713-3760

Verbatim source input

```
maximize (20),
X Z
w(01 , 02 )d(01 , 02 )
(28)
0≤i<j<M

02 ∈Si ,01 ∈Sj

are {si = Mi }M
i=0 , i.e. the same as those for Kendall's τ .
Proof.
X
0≤i<j<M

Z
w(01 , 02 )d(01 , 02 ) =
02 ∈Si ,01 ∈Sj

Z
X

(01 - 02 )d(01 , 02 )
0≤i<j<M

=

02 ∈Si ,01 ∈Sj

[U+0012 control character]

X
0<i<j≤M

sj + sj-1 si + si-1
-
2
2

[U+0013 control character]
(si - si-1 )(sj - sj-1 )

Finding an asymptotically optimal {si } then is a constrained third order polynomial maximization
i=M , as for Kendall's tau
problem with M variables. The maximum is achieved at {si = Mi }i=0
correlation.
```

Expanded PaperInterface specification

```
∀ {M : ℕ} [Nonempty (Fin M)] (s : ℕ → ℝ),
  Monotone s →
  s 0 = 0 →
  s M = 1 →
  GJ190ptimalBinaryRatingSystems.spearmanLinearWeightOrderedPairIntervalObjective M s ≤ (1 - (↑M)-1 ^ 2) / 6
```

Lean proof endpoint (built separately)

GJ190ptimalBinaryRatingSystems.PaperInterface.paper_lemmaC12_spearman_linear_weight_ordered_pair_interval_objective_le_equispaced

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation